PRAYAS

JEE 2026

Physics

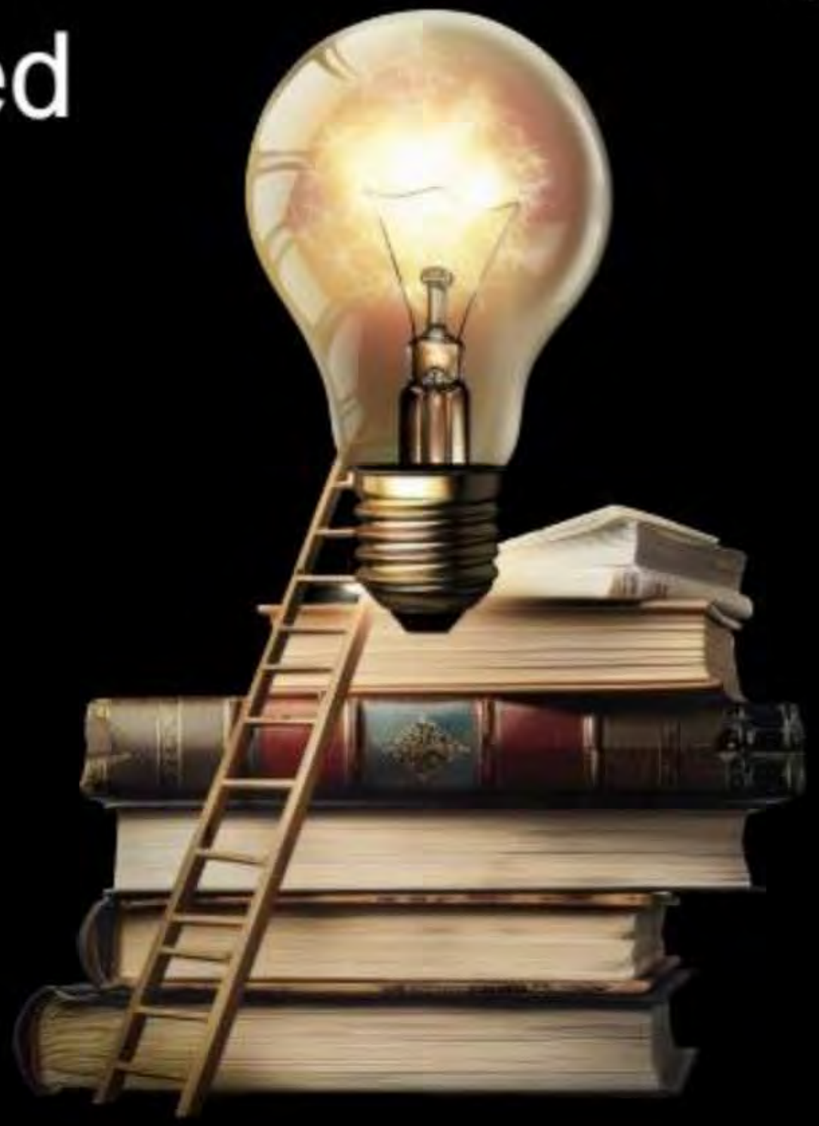
Electromagnetic Induction

Lecture – 3

Manish Singh Tak (Masti Sir)

Topics to be covered

1. Electromagnetic Induction
2.
3.
4.
5.

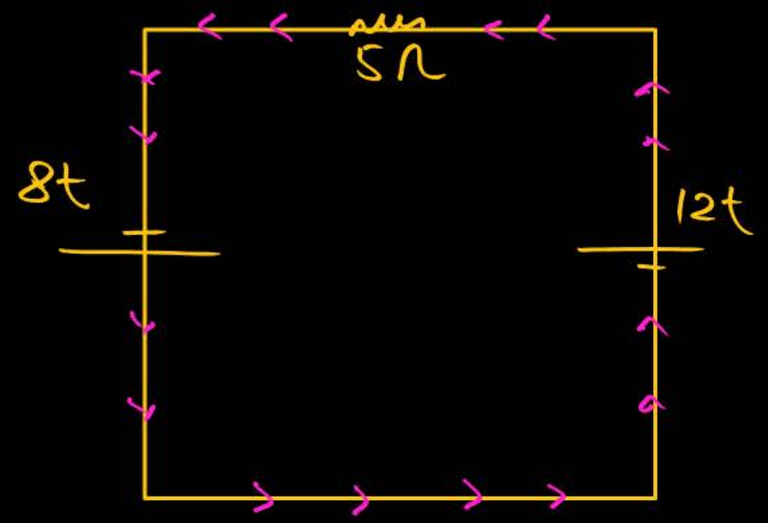


Question

A square loop of side 2m and resistance 5Ω is placed in uniform magnetic field as shown. Find current induce in the loop at time t (t = 1 Sec.)

$$\mathcal{E}.m.f_1 = -\frac{d}{dt}\left(2(2t^2)\right) = -8t$$

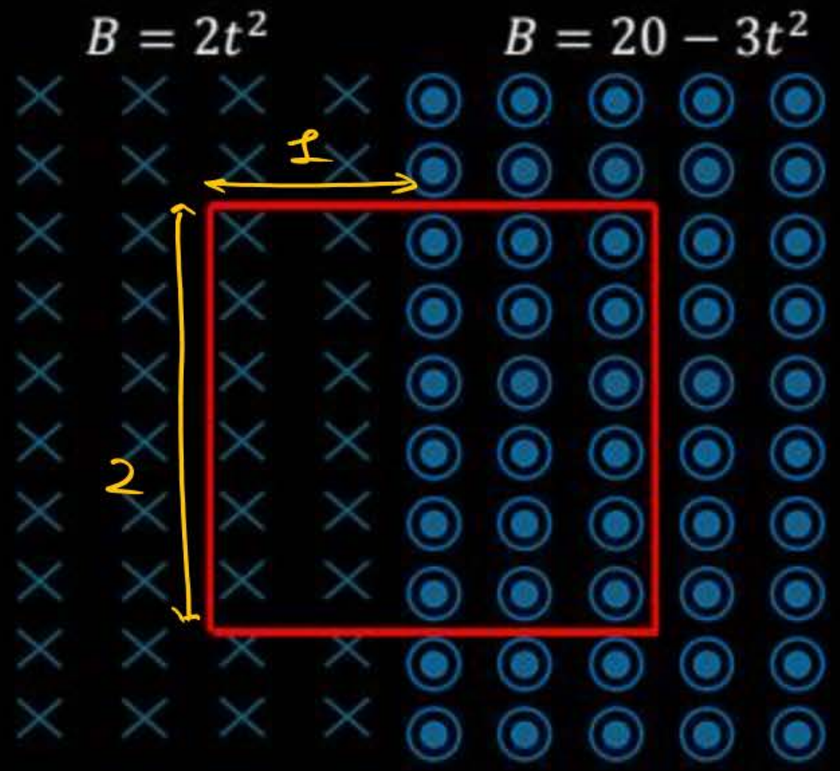
$$\mathcal{E}.m.f_2 = -\frac{d}{dt}\left(2(20-3t^2)\right) = 12t$$



$$i = \frac{12t+8t}{5}$$

$$i = 4t$$

$$t=1 \quad i = 4A$$



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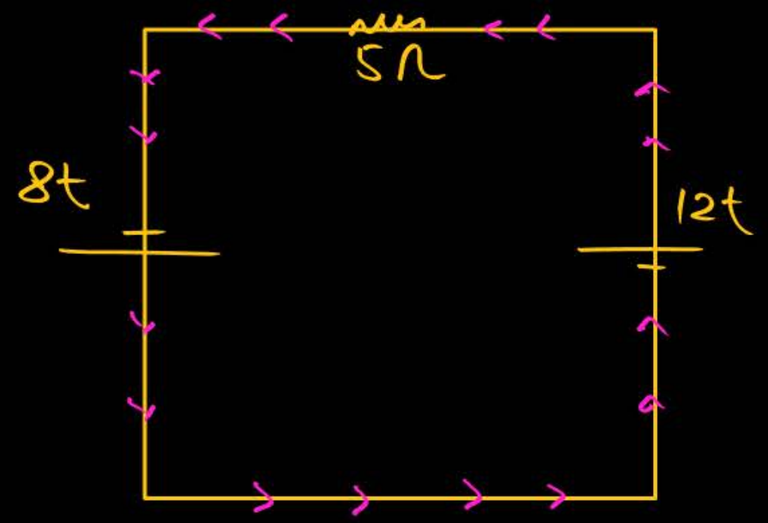
$$\mathcal{E}.m.f_1 = -\frac{d}{dt}\left(2(2t^2)\right) = -8t$$

$$\mathcal{E}.m.f_2 = -\frac{d}{dt}\left(-2(20-3t^2)\right) = -12t$$



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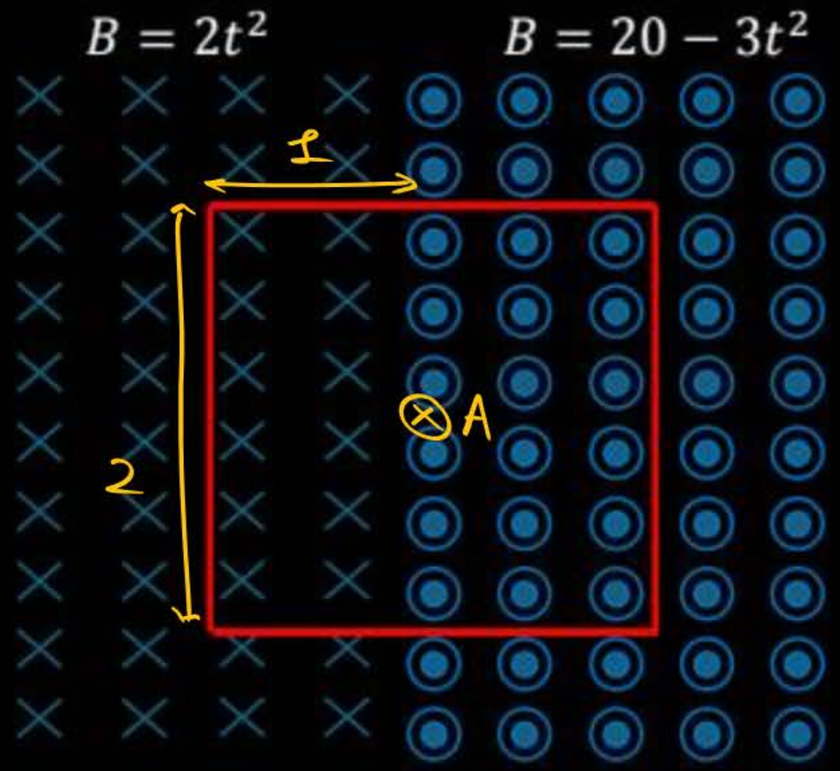
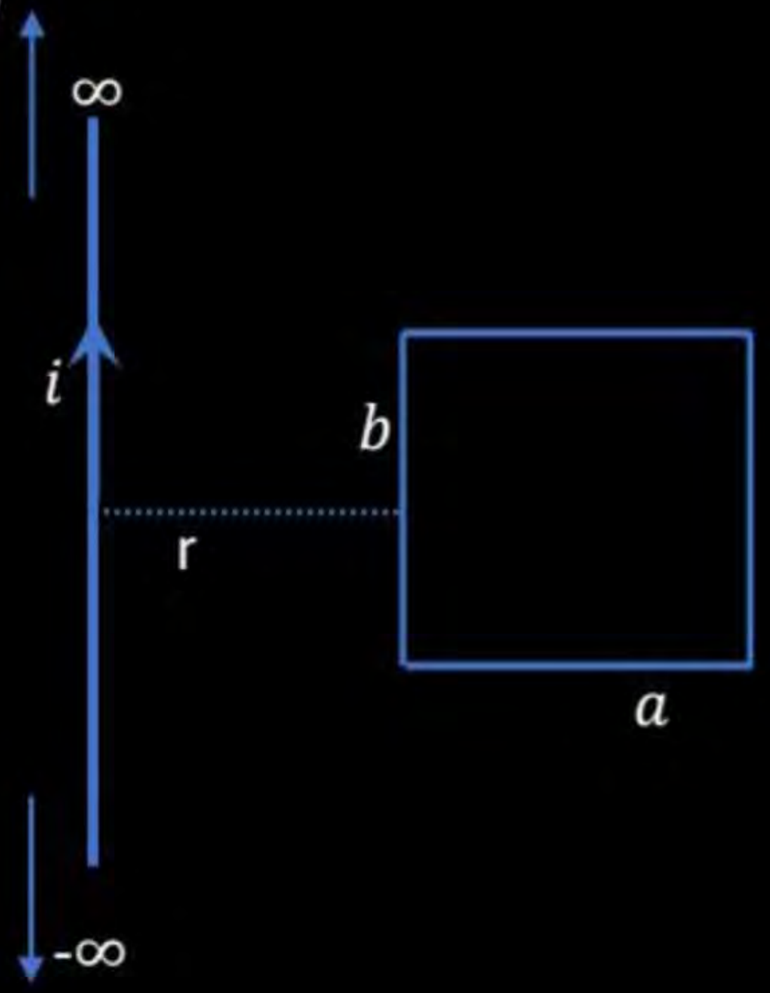


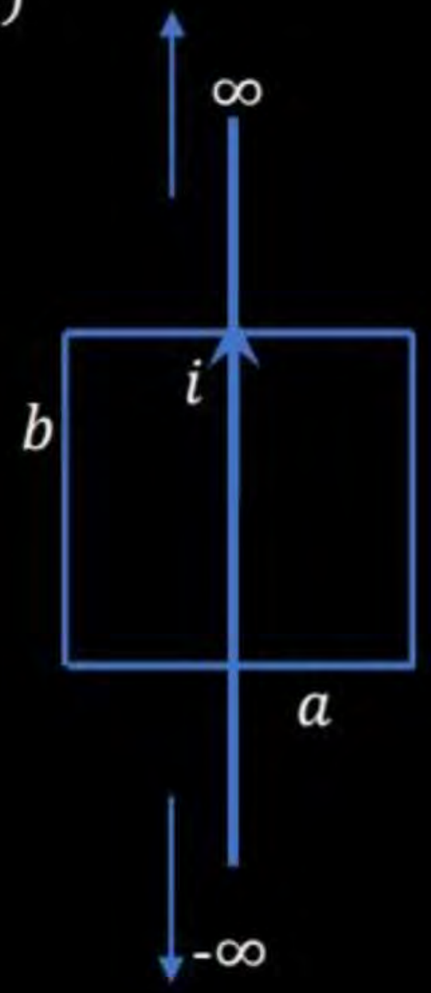
Question

Find Current induce in rectangular loop of resistance R due to very long current carrying straight wire. Current in Wire is i = $2t^3$ A

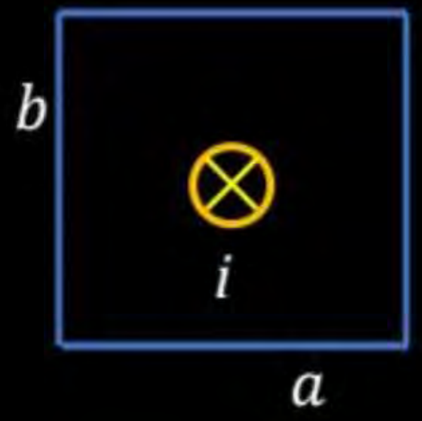
(1)



(2)



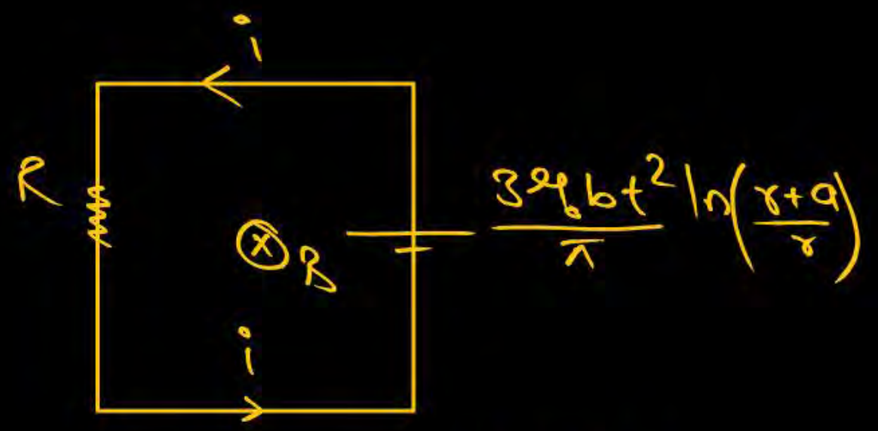
(3)



$$\phi = \frac{\mu_0 i b}{2\pi} \ln\left(\frac{r+a}{r}\right)$$

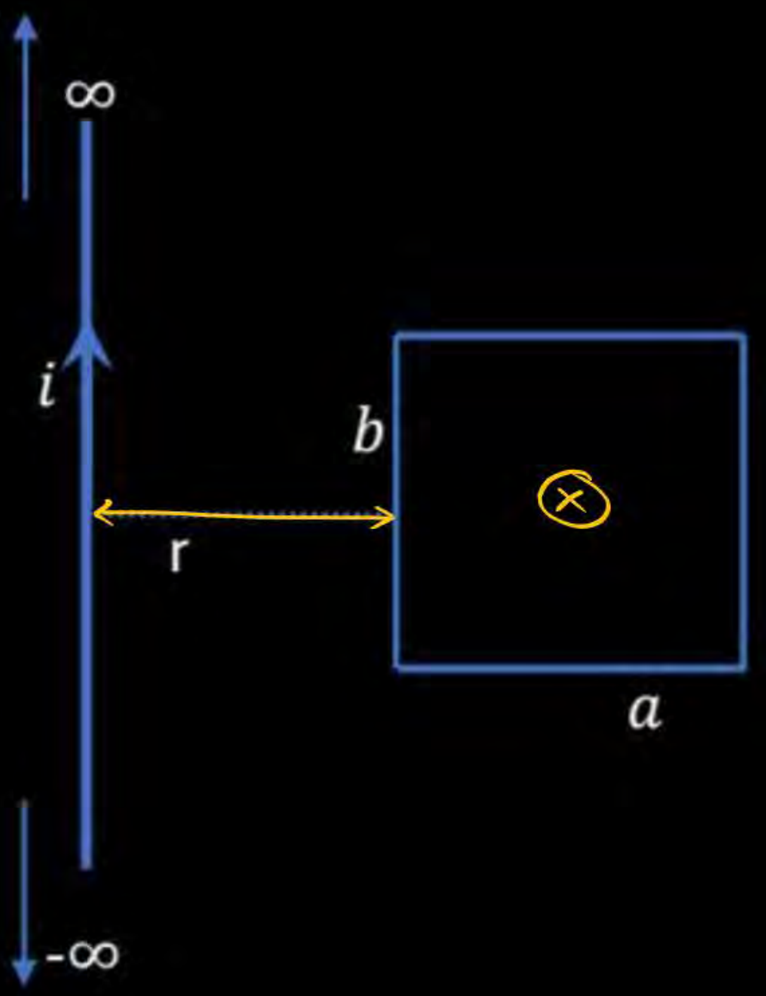
$$\phi = \frac{\mu_0 (2t^3) b}{2\pi} \ln\left(\frac{r+a}{r}\right)$$

$$\mathcal{E}.m.f. = -\frac{d\phi}{dt} = -\frac{3\mu_0 b t^2}{\pi} \ln\left(\frac{r+a}{r}\right)$$



ϕ increases with time

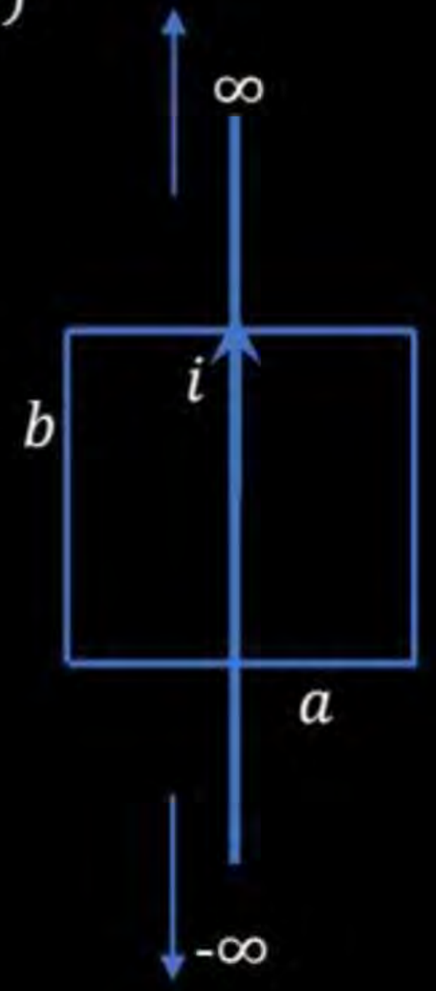
(1)



$\phi = 0$ for any value of i, all the time

(2)

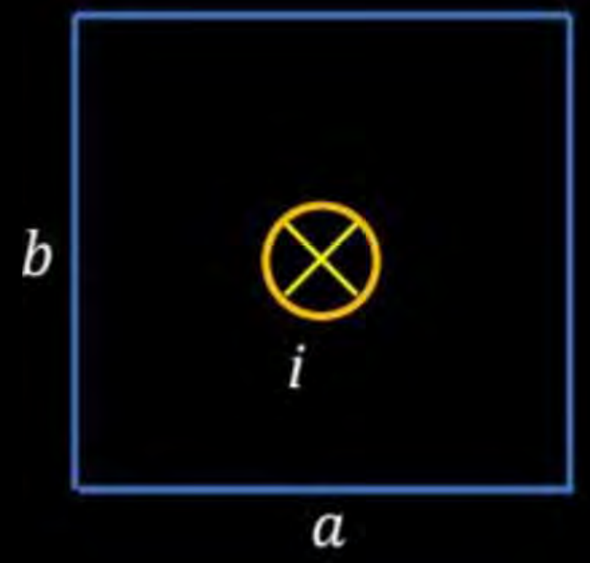
$\mathcal{E}.m.f. = -\frac{d}{dt}(0) = 0$



$\phi = 0$ for any value of i, at any time t

(3)

$\mathcal{E}.m.f. = 0$

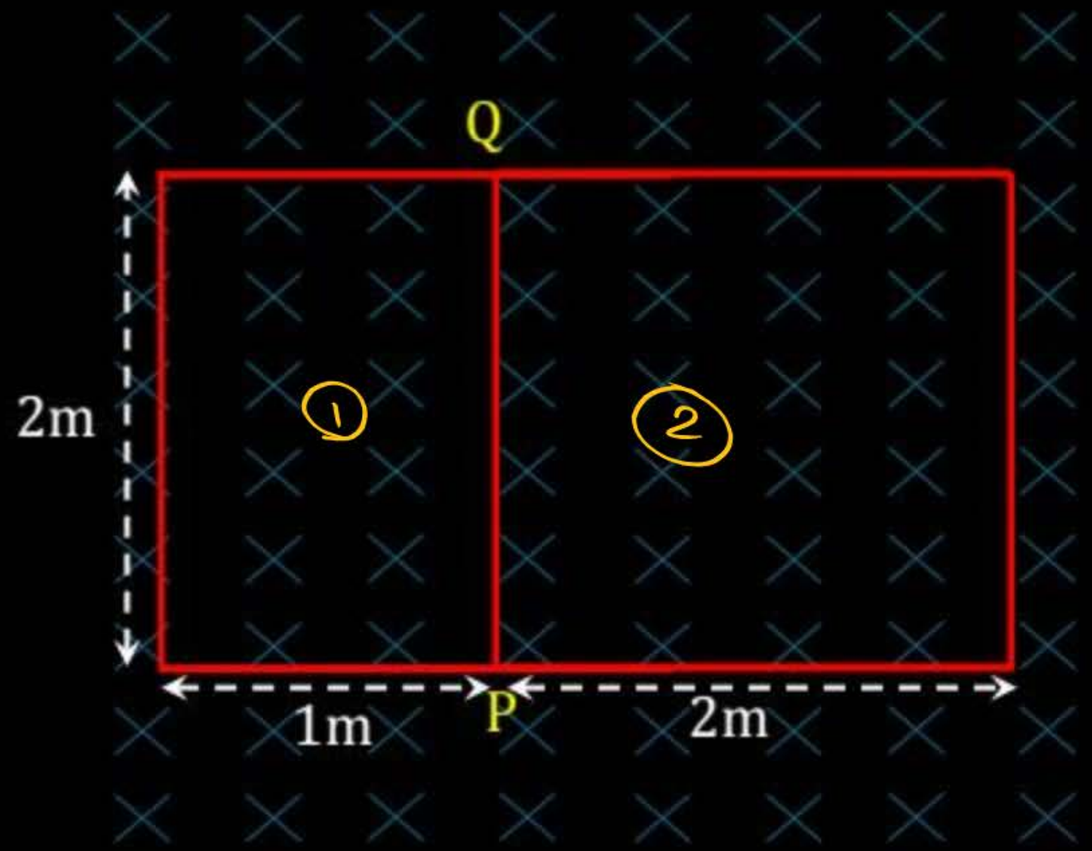


Question

ans. $(i_{PQ})110 =$

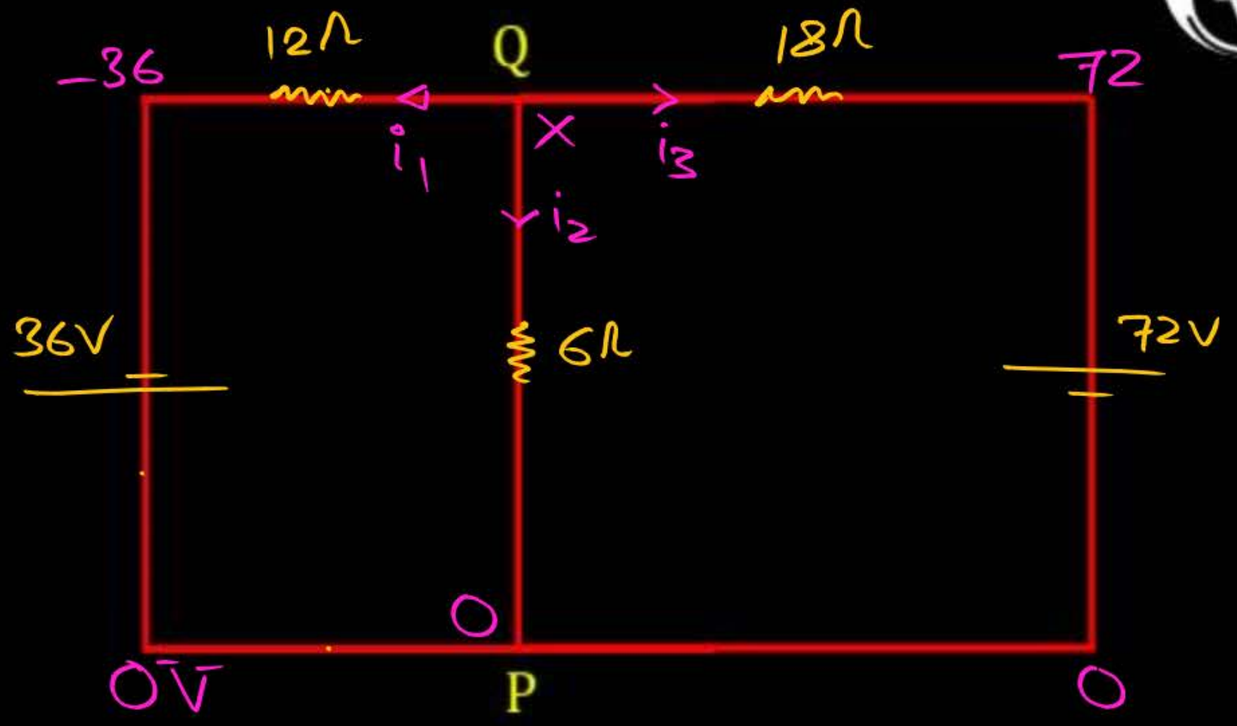
Given loop have resistance $3\Omega/\text{m}$ and it is placed in uniform magnetic field $B = 18t$ T as shown. Find current induce in the side PQ

$\mathcal{E}.m.f_1$



$$\mathcal{E}.m.f_1 = -\frac{d}{dt}(2\times 18t) = -36\ V$$

$$\mathcal{E}.m.f_2 = -\frac{d}{dt}(4\times 18t) = -72\ V$$



$$\frac{X-(-36)}{12} + \frac{X-0}{6} + \frac{X-72}{18} = 0$$

$$\frac{3X+108+6X+2X-144}{36} = 0$$

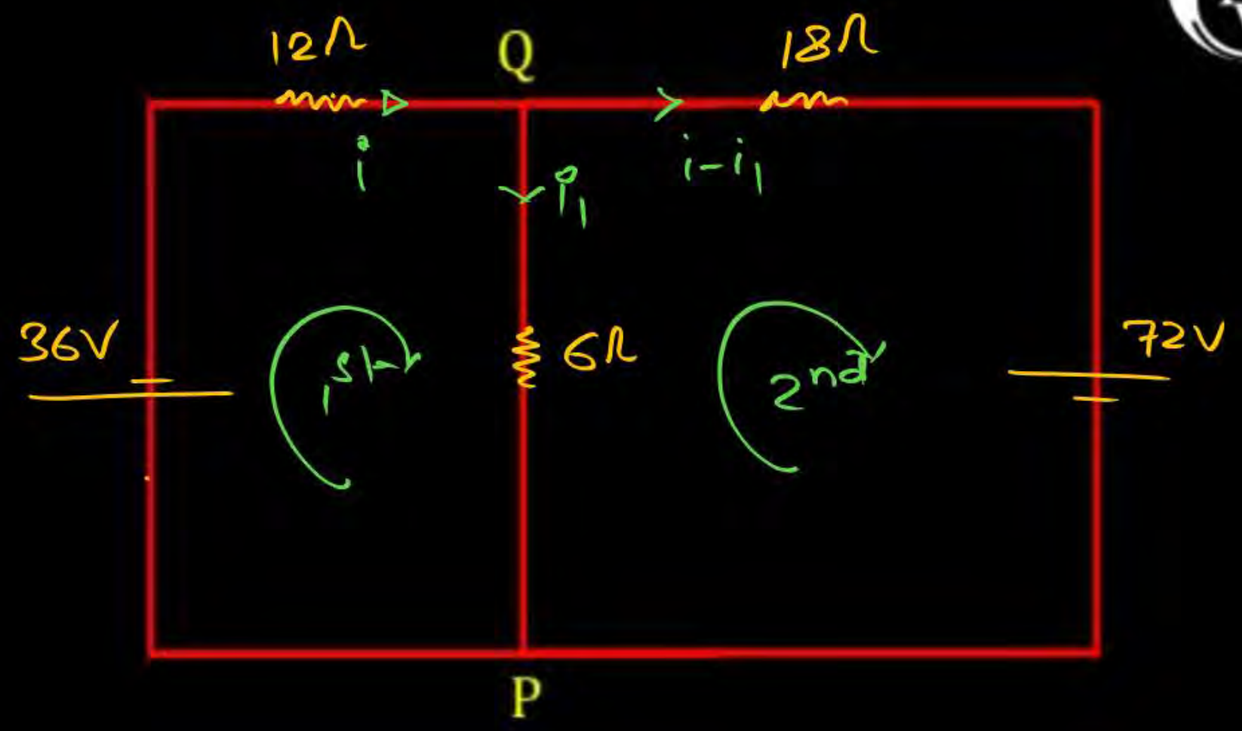
$$X = \frac{36}{11}\ \text{Volt}$$

$$i_2 = \frac{\frac{36}{11}-0}{6} = \frac{6}{11}\ A$$

Ans $\left(110\, i_2 = 110\times\frac{6}{11} = 60\right)$

1st $\quad -36 - 12i - 6i_1 = 0$

2nd $\quad 6i_1 - 18(i - i_1) - 72 = 0$

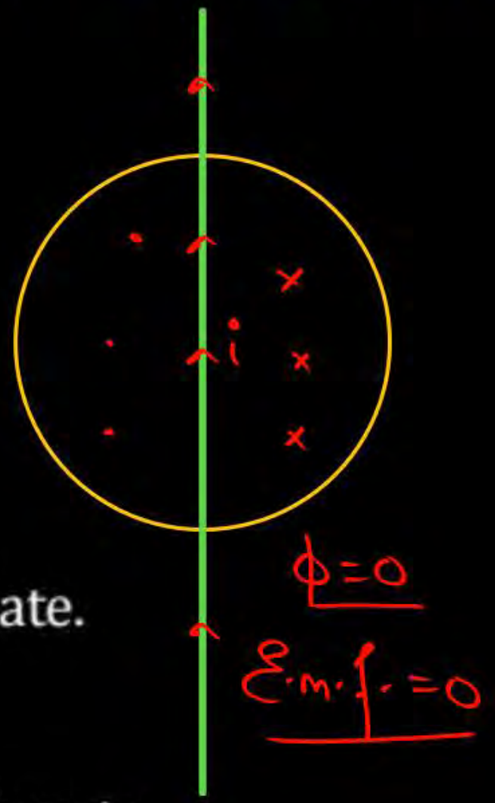


Question

AC or ABC

A current carrying infinitely long wire is kept along the diameter of a circular wire loop, without touching it, the correct statement(s) is(are) [Jee Adv. 2012]

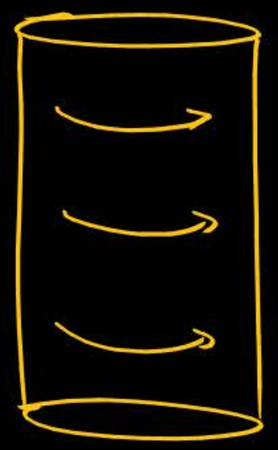
1. The emf induced in the loop is zero if the current is constant.
2. The emf induced in the loop is finite if the current is constant.
3. The emf induced in the loop is zero if the current decreases at a steady rate.
4. The emf induced in the loop is infinite if the current decreases at a steady rate.



Question

$(\pi^2 \approx 10)$

A long circular tube of length 10 m and radius 0.3 m carries a current I along its curved surface as shown. A wire-loop of resistance 0.005 ohm and of radius 0.1 m is placed inside the tube with its axis coinciding with the axis of the tube. The current varies as $I = I_0\cos(300t)$ where I_0 is constant. If the magnetic moment of the loop is $N\mu_0 I_0 \sin(300t)$, then ' N ' is [Jee Adv. 2011]



Solenoid

$$B = \frac{\mu_0 N i}{l} = \frac{\mu_0 (1) i}{10}$$

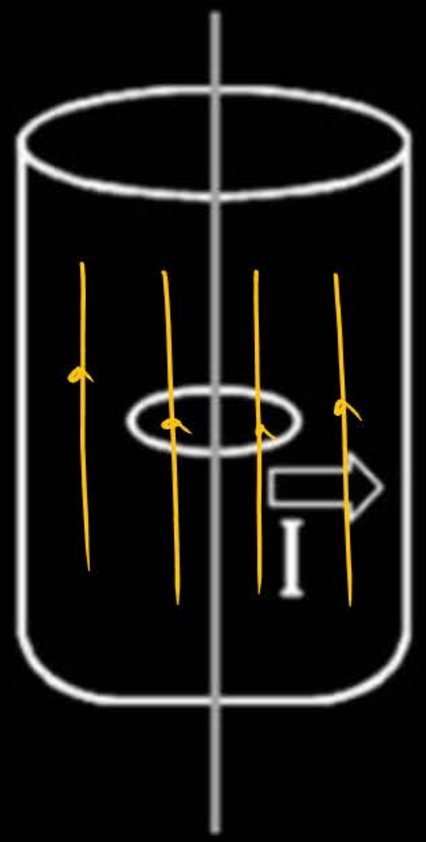
$$\phi = BA = \left(\frac{\mu_0 i}{10}\right)\pi r^2$$

$$\text{E.m.f.} = -\frac{d\phi}{dt} = -\frac{\mu_0 \pi r^2}{10}\frac{di}{dt}$$

$$= \frac{\mu_0 \pi r^2}{10}(300 i_0 \sin 300t)$$

$$i = \frac{\text{E.m.f}}{R}$$

$$M = i \cdot A = \frac{\mu_0 \pi r^2}{10R}(300 i_0 \sin 300t) \cdot \pi r^2$$



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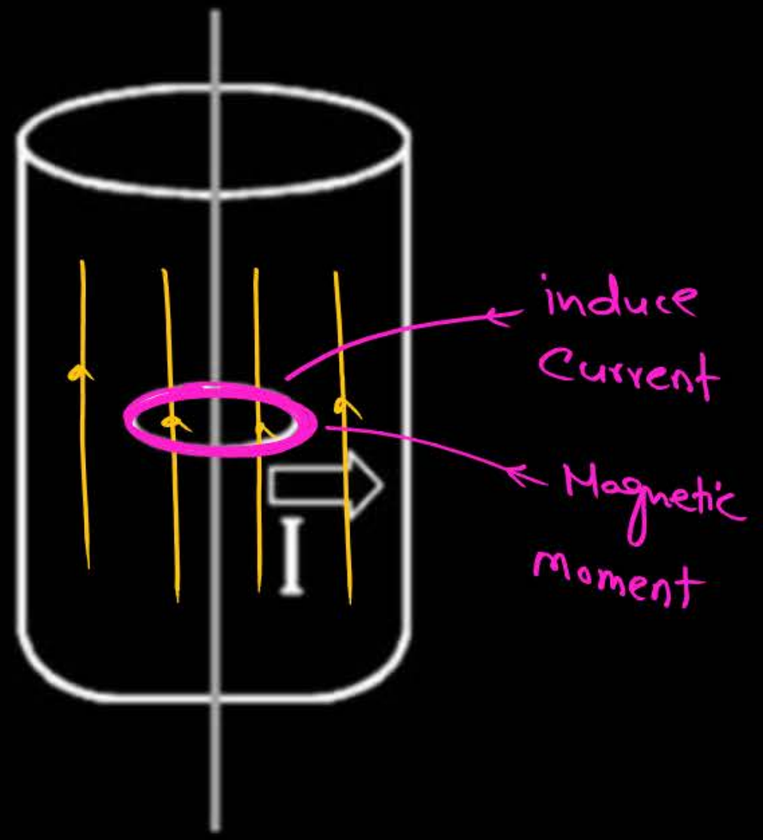
$$M = (\mu_0 i_0 \sin 300t)\left[\frac{\pi^2 r^4 \cdot 300}{10R}\right]$$

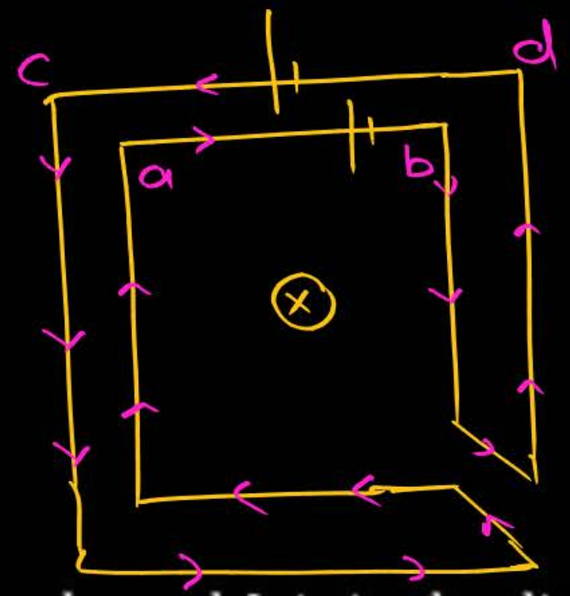
$$\frac{(10)(10^{-1})^4 \times 300}{10 \times 5 \times 10^{-3}} = \frac{300}{50} = 6$$



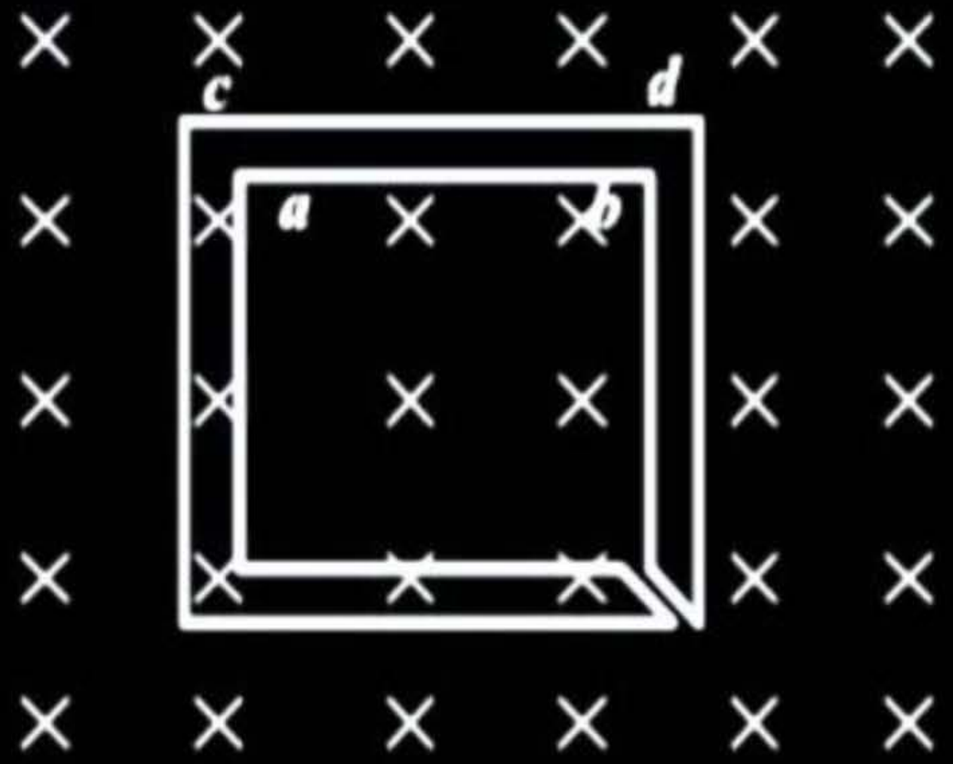
Question

The figure shows certain wire segments joined together to form a coplanar loop. The loop is placed in a perpendicular magnetic field in the direction going into the plane of the figure. The magnitude of the field increases with time. I_1 and I_2 are the currents in the segment's ab and cd. Then, [Jee Adv. 2009]

B increase , ϕ increase



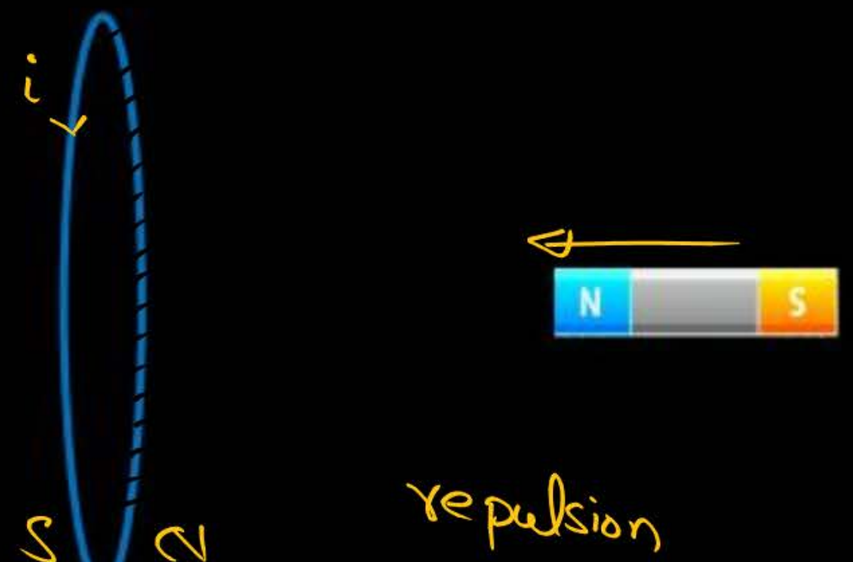
$|i_1| = |i_2|$



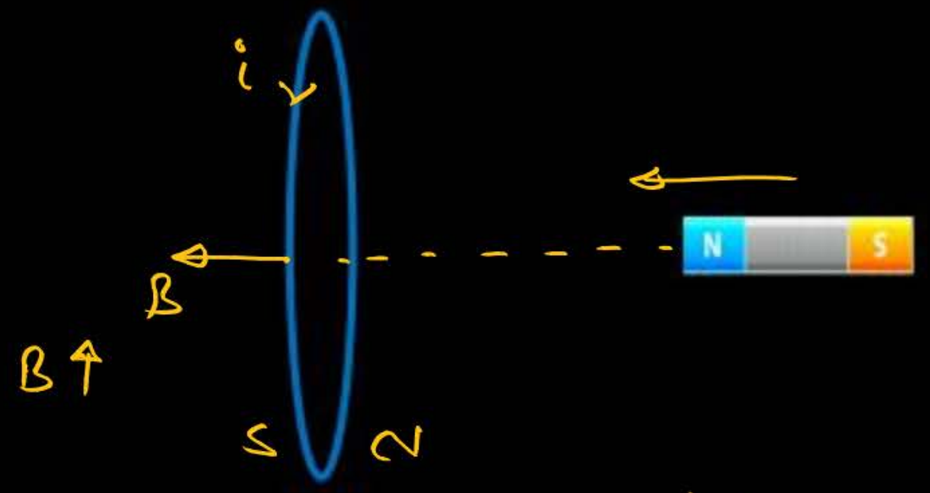
1. $I_1 > I_2$
2. $I_1 < I_2$
3. I_1 is in the direction ba and I_2 is in the direction cd
4. I_1 is in the direction ab and I_2 is in the direction dc

Lenz's Law

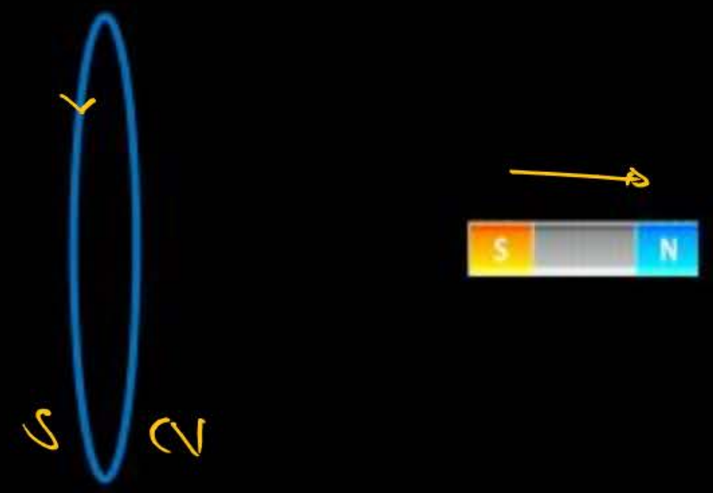
Electro-magnetic induction occurs in such a way that it will oppose the cause by which it is generated

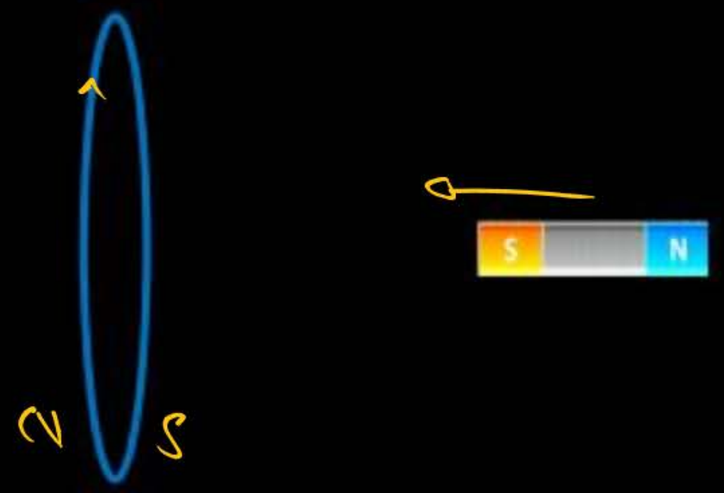


attraction



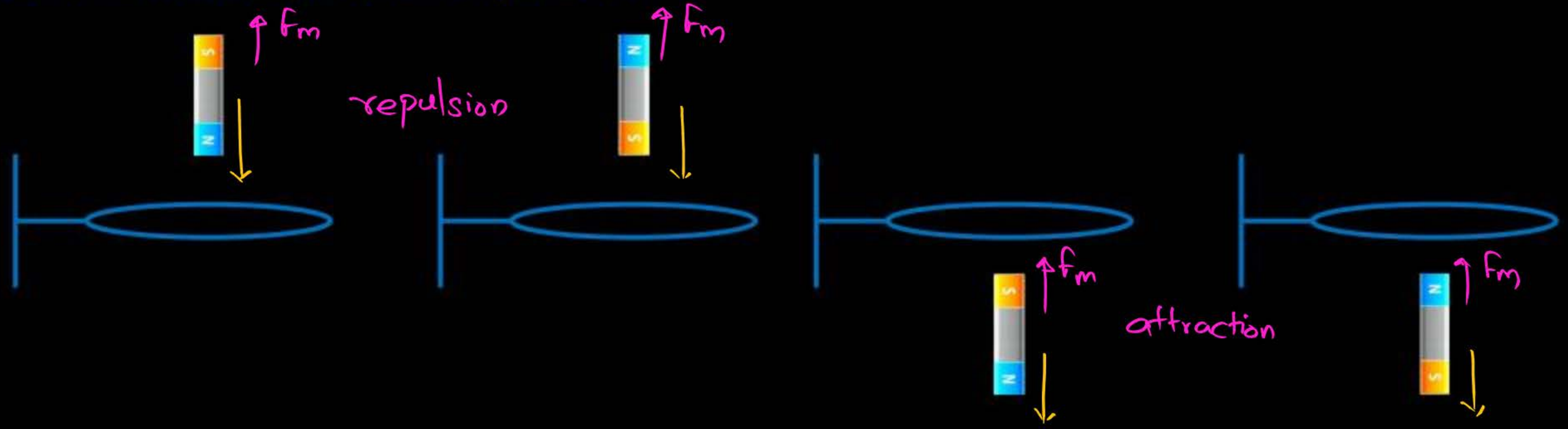
repulsion





Question

A bar magnet is released from rest on axis of fixed conducting loop as shown, after dt time acceleration of magnet is a. compare a with g



Question

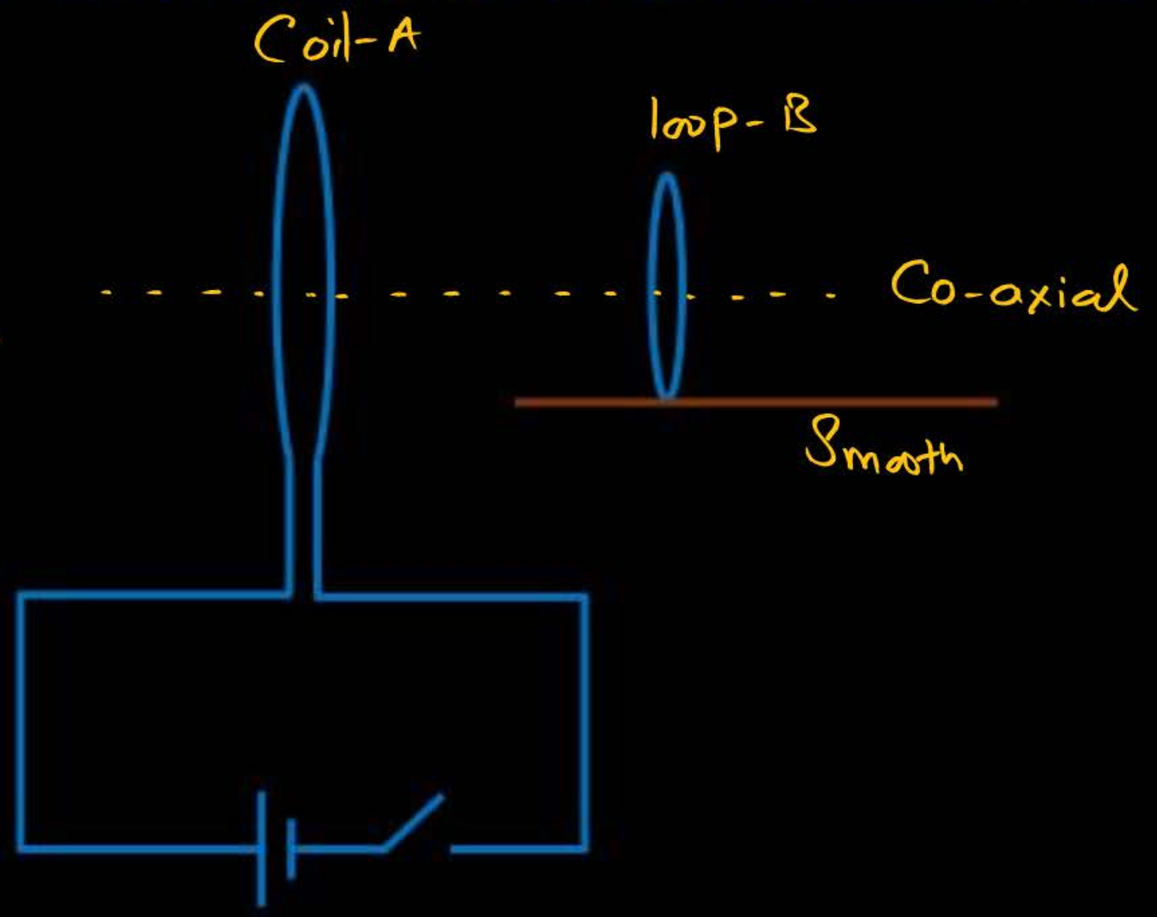
A conducting loop is kept on smooth horizontal surface as shown. Find in which direction loop tends to move if (1) Initially switch was opened and now it is close (2) Initially switch was closed and now it is open

(1) initially no Current in Coil A So no flux in loop B

Now when Switch is closed, Current flow in Coil A ⇒ flux in loop B

पहले ϕ नहीं था, अब ϕ आया तो loop कम ϕ की तरफ भागेगा (repulsion)

$(+\hat{i})$



Question

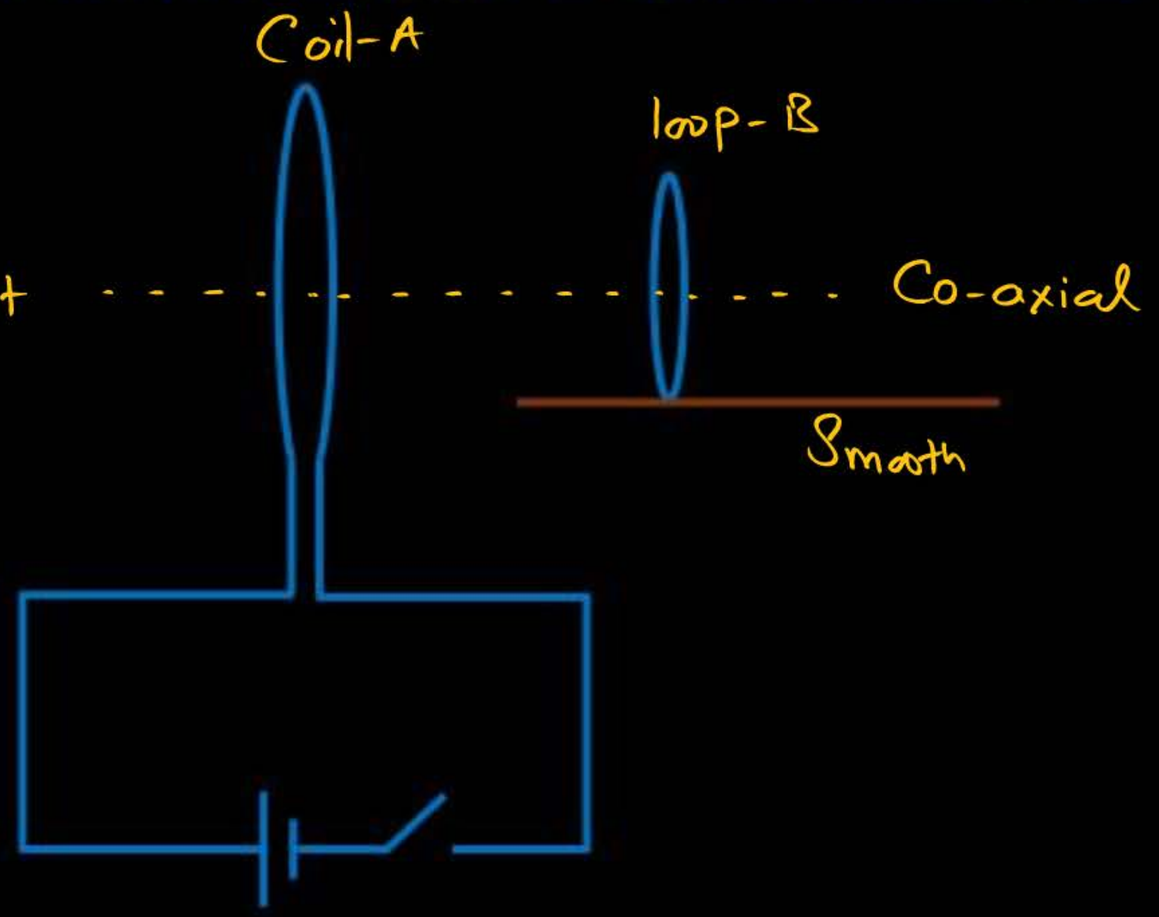
A conducting loop is kept on smooth horizontal surface as shown. Find in which direction loop tends to move if (1) Initially switch was opened and now it is close (2) Initially switch was closed and now it is open

(2) initially Current in Coil A
$\Rightarrow \phi$ in loop B

Now when Switch is open, Current in Coil-A dies $\Rightarrow \phi$ dies
(in few ms)

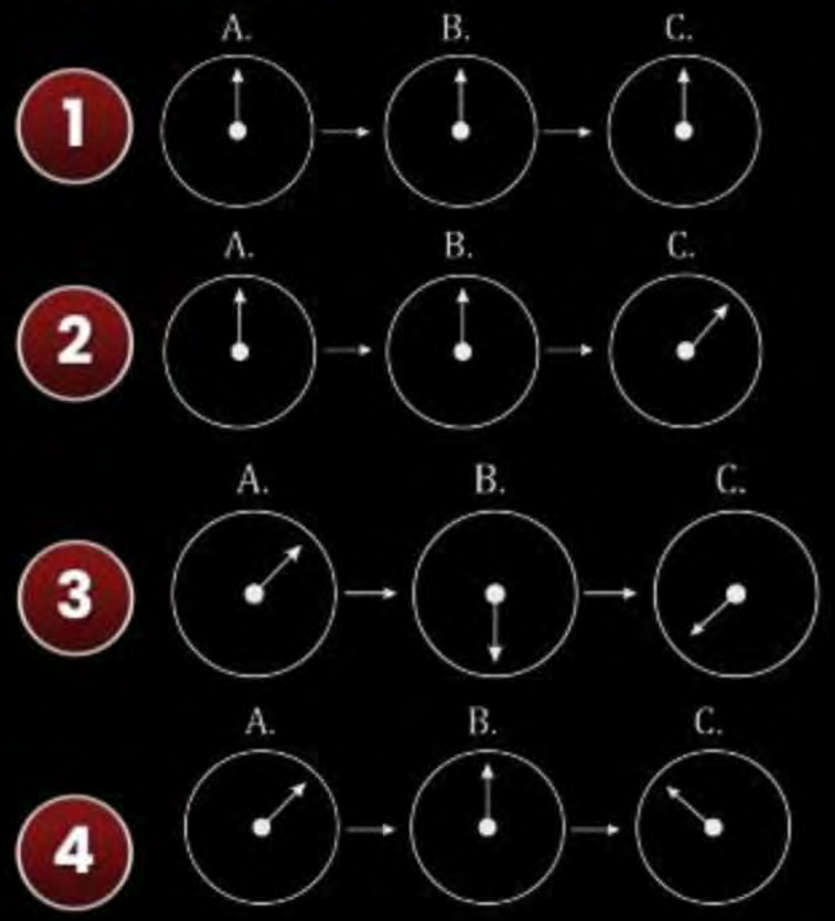
पहले ϕ था, अब ϕ नहीं है तो loop ज्यादा ϕ की तरफ भागेगा (attraction)

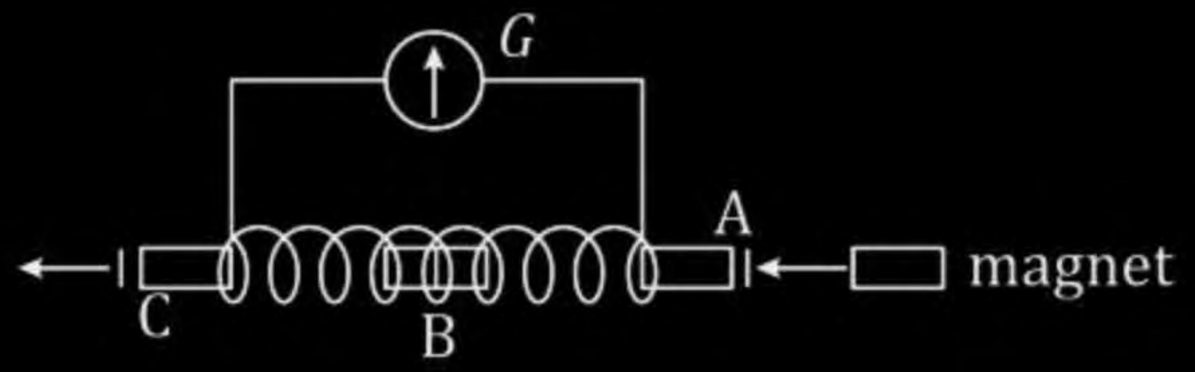
$-\hat{i}$



Question

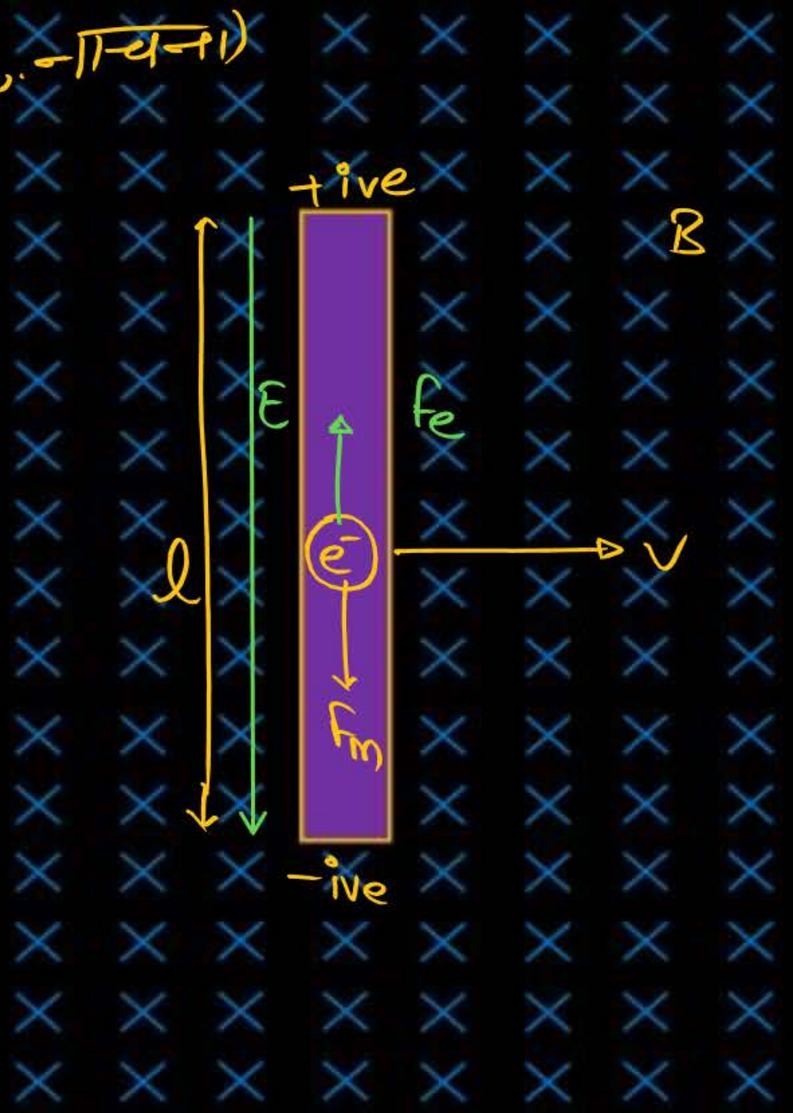
A small bar magnet is moved through a coil at constant speed from one end to the other. Which of the following series of observations will be seen on the galvanometer G attached across the coil. Three positions shown describe: (1) the magnet's entry (2) magnet is completely inside and (3) magnet's exit.





Motional EMF

When a Conducting wire move (चलना, घुमना, गिरना) in magnetic field and magnetic lines are cut by the wire then an e.m.f. is induce in the wire (known as motional e.m.f.)



At Steady State :→ $\vec{F}_e = -\vec{F}_m$

$$(-e)\vec{E} = -\left((-e)(\vec{v}\times\vec{B})\right) \Rightarrow \vec{E} = -(\vec{v}\times\vec{B})$$

Voltage (e.m.f produce) across Rod

$$e.m.f. = -\int \vec{E}\cdot d\vec{l} \Rightarrow \boxed{e.m.f. = \int (\vec{v}\times\vec{B})\cdot d\vec{l}}$$

$$\mathcal{E}.m.f = \int (\vec{V} \times \vec{B}) \cdot d\vec{l}$$

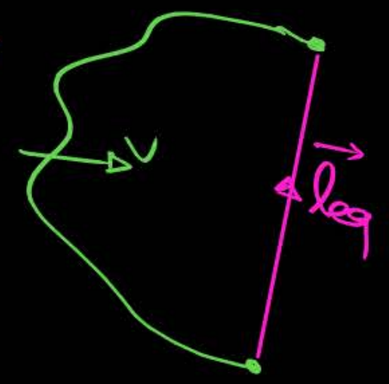
for translatory motion of wire in uniform magnetic field

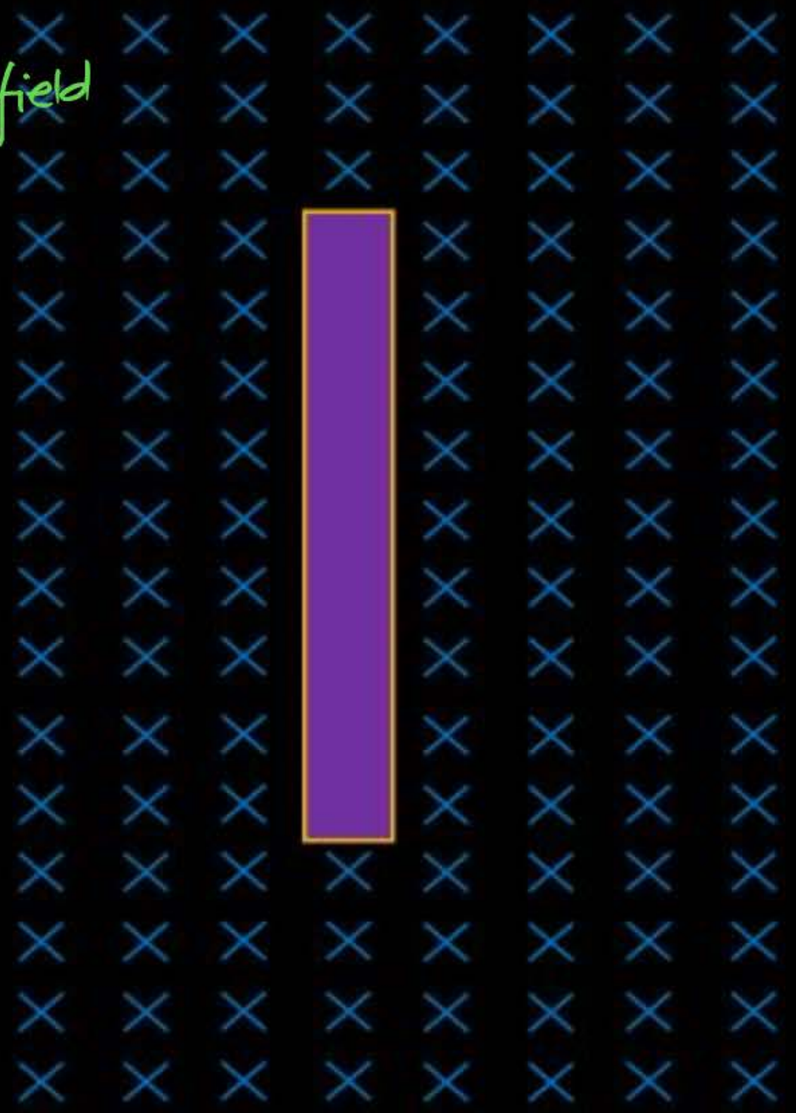
$$\mathcal{E}.m.f. = (\vec{V} \times \vec{B}) \cdot \int d\vec{l}$$

$$\boxed{\mathcal{E}.m.f. = (\vec{V} \times \vec{B}) \cdot \vec{l_{eq}}} = [\vec{V} \ \vec{B} \ \vec{l_{eq}}]$$

$\vec{l_{eq}} \rightarrow$ Equivalent length vector

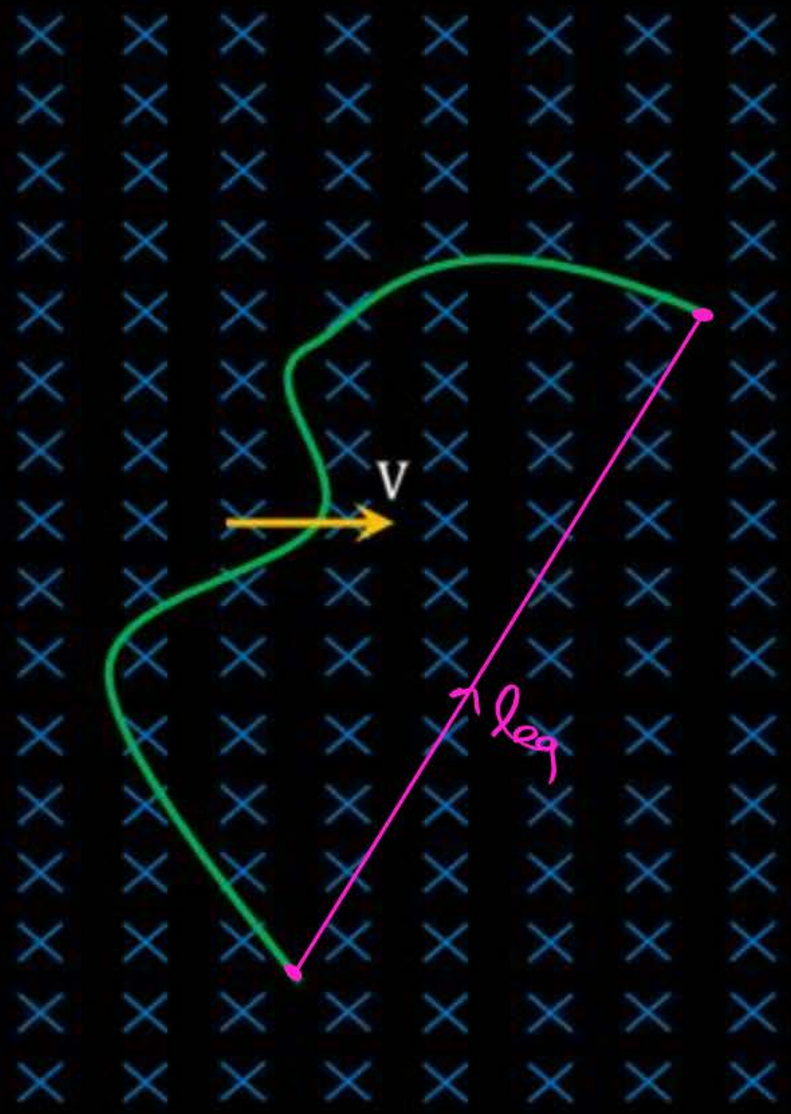
(vector joining ends of wire)





$\int dl$

$\int d\vec{l}$


v
leq

$$\mathcal{E}.m.f. = (\vec{V} \times \vec{B}) \cdot \vec{l_{eq}} = \vec{l_{eq}} \cdot (\vec{V} \times \vec{B})$$

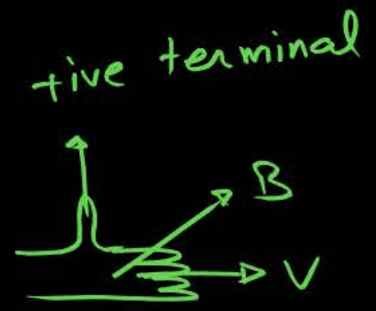
$\mathcal{E}.m.f. = 0$ if all three vector $\vec{V}$, $\vec{B}$ and $\vec{l_{eq}}$ are in parallel

$\mathcal{E}.m.f. = 0$ if any two vectors are parallel

$\mathcal{E}.m.f = (\vec{v} \times \vec{B}) \times \vec{l}_{eq}$ $\quad \vec{v} \perp \vec{B} \perp \vec{l}$

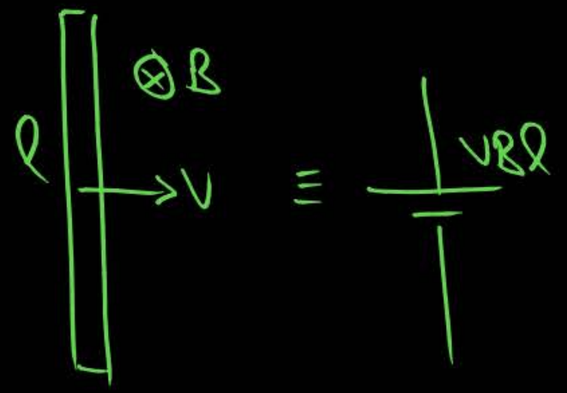
** $\boxed{\mathcal{E}.m.f = vBl}$

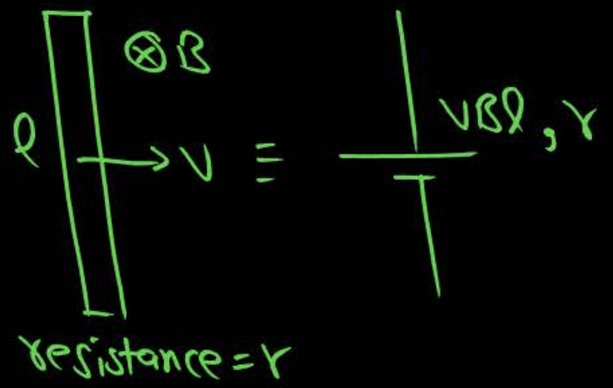
+ive terminal of battery
$\Rightarrow$ in dir^n of $\vec{v} \times \vec{B}$

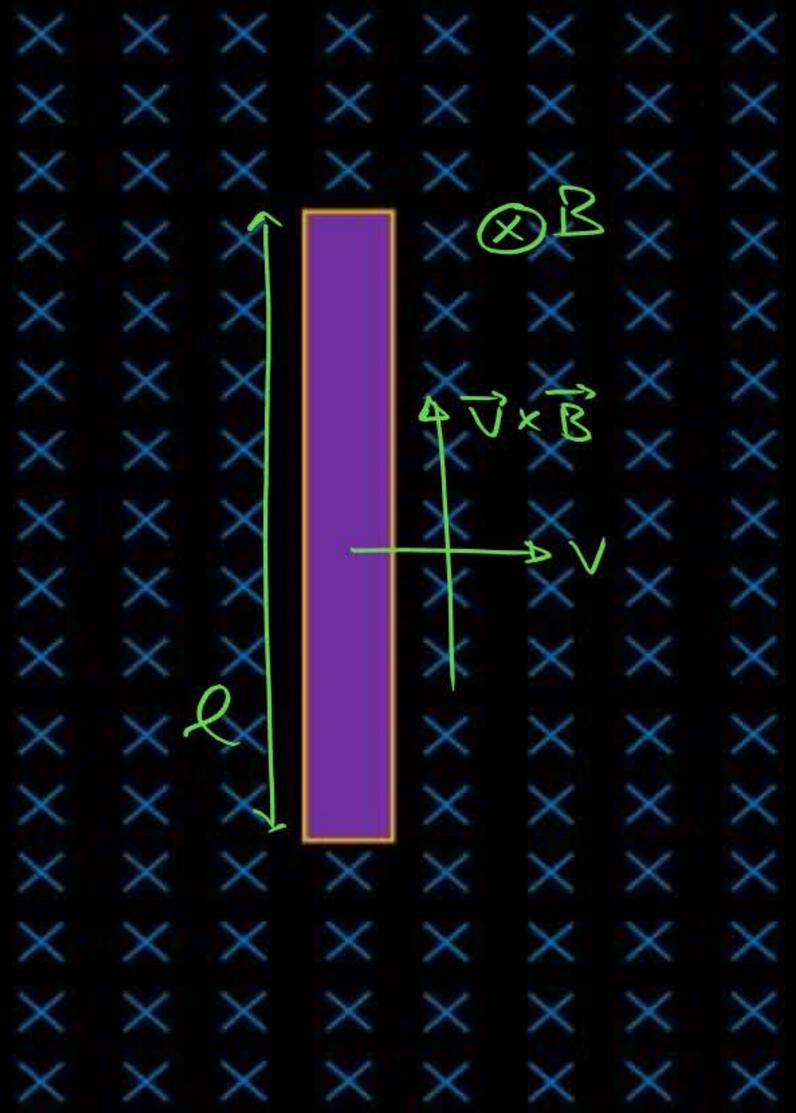


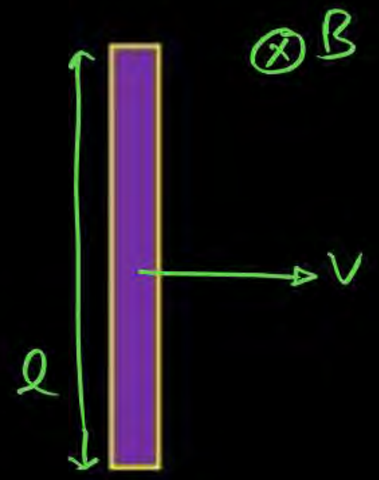
V → अंगुलियां , B → Palm

अंगूठा ⟶ +ive terminal









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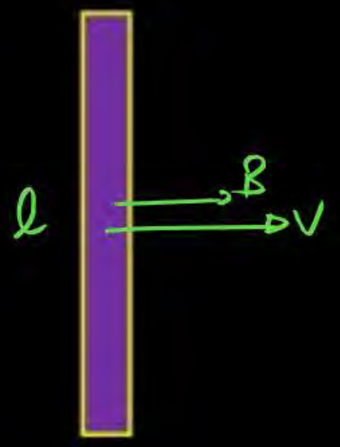
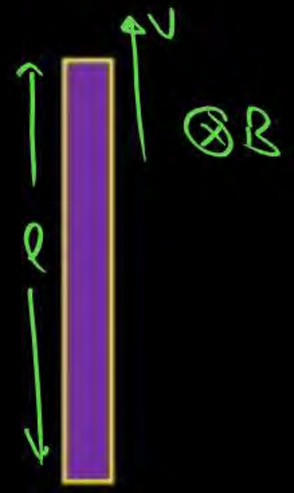
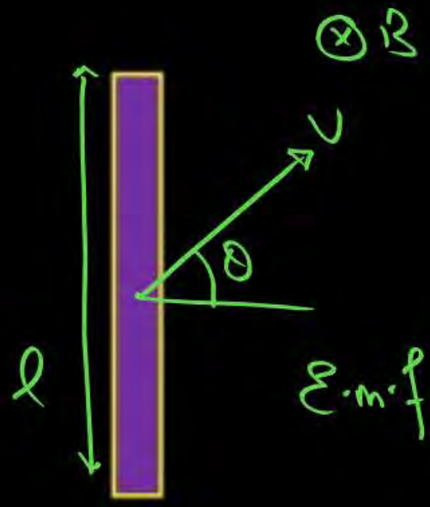


$\mathcal{E}.m.f = 0$

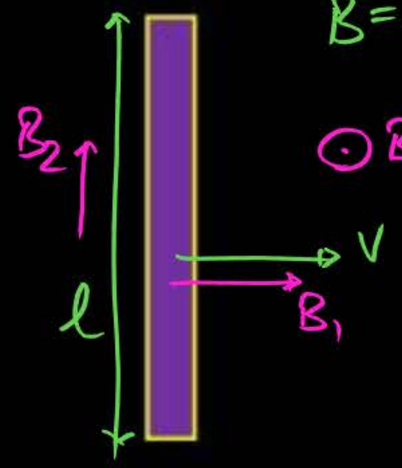


$\mathcal{E}.m.f_{Rod} = 0$

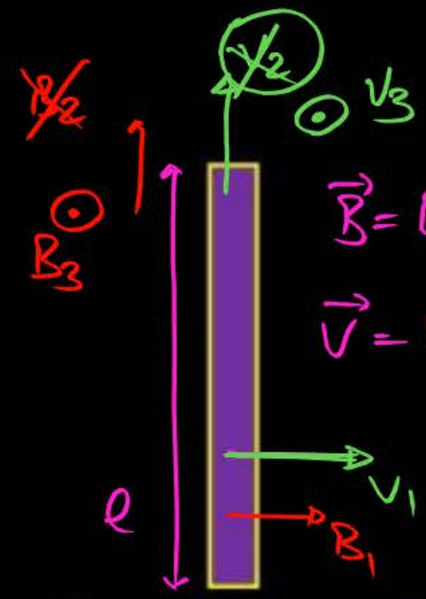


$\mathcal{E}.m.f = (V\cos\theta)(Bl)$

$\vec{B} = B_1\hat{i} + B_2\hat{j} + B_3\hat{k}$



$\mathcal{E}.m.f = v B_3 l$

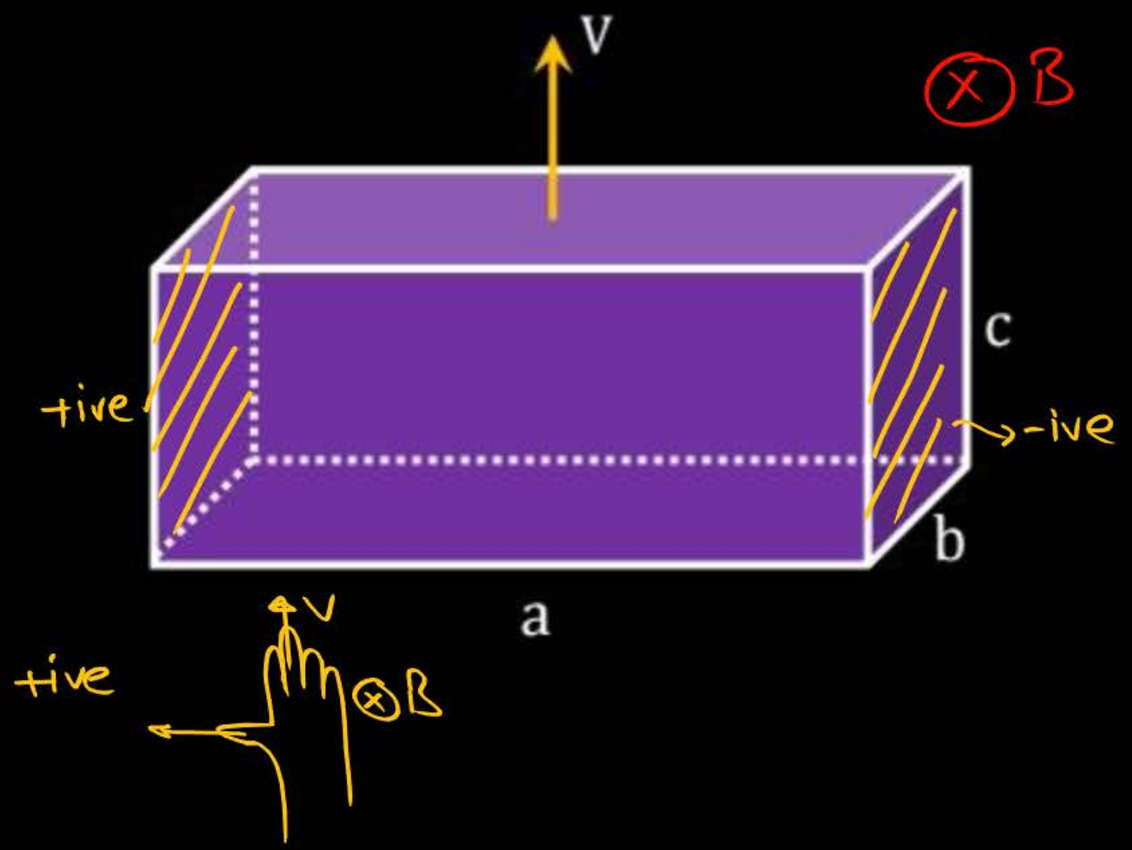


$\vec{B} = B_1\hat{i} + B_2\hat{j} + B_3\hat{k}$

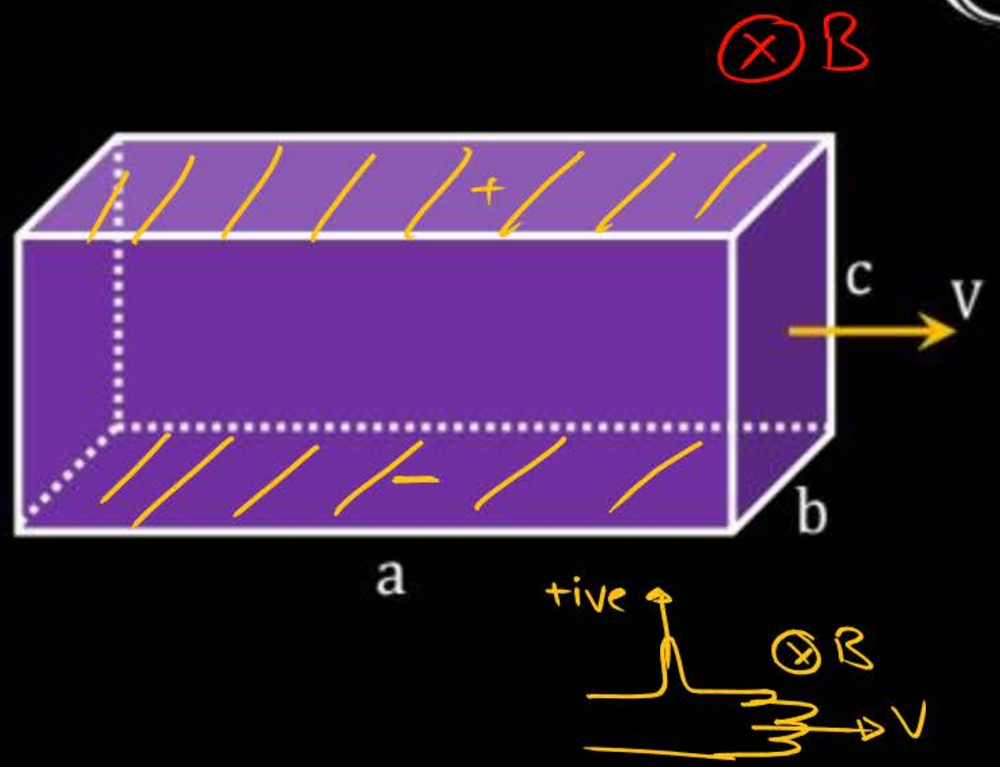
$\vec{v} = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}$

$B_3 - v_1 - l \longrightarrow B_3 v_1 l$

$B_1 - v_3 - l \longrightarrow B_1 v_3 l$



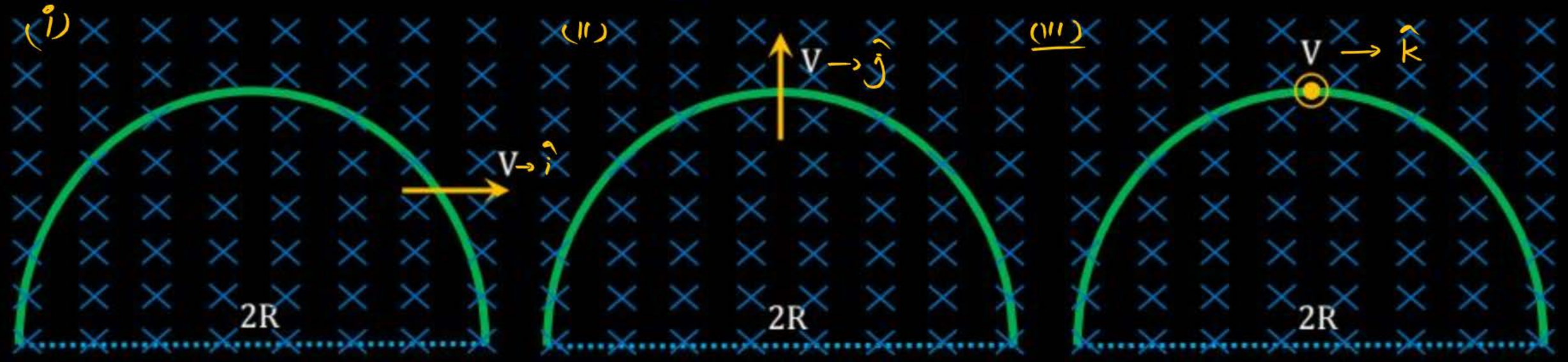
$\mathcal{E}.m.f = vBa$



$\mathcal{E}.m.f. = vBc$

Question

A thin semicircular conducting ring of radius R is moving with velocity v in uniform magnetic field B as shown. Find emf induce in the ring



$l_{eq} = 2R$ ⊗B

→V

$\mathcal{E}.m.f_{Red} = 0$

V ⊗B

2R

$\mathcal{E}.m.f. = VB(2R)$

⊙V ⊗B

2R

$\mathcal{E}.m.f = 0$

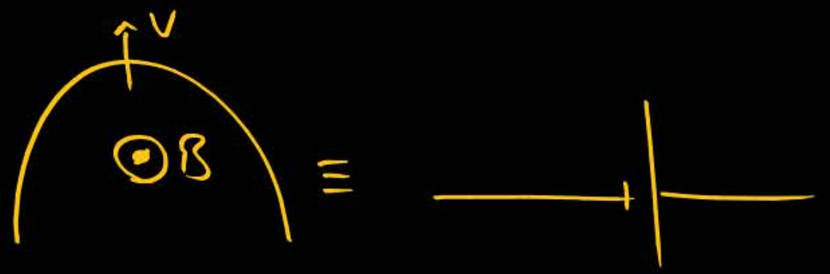
Question

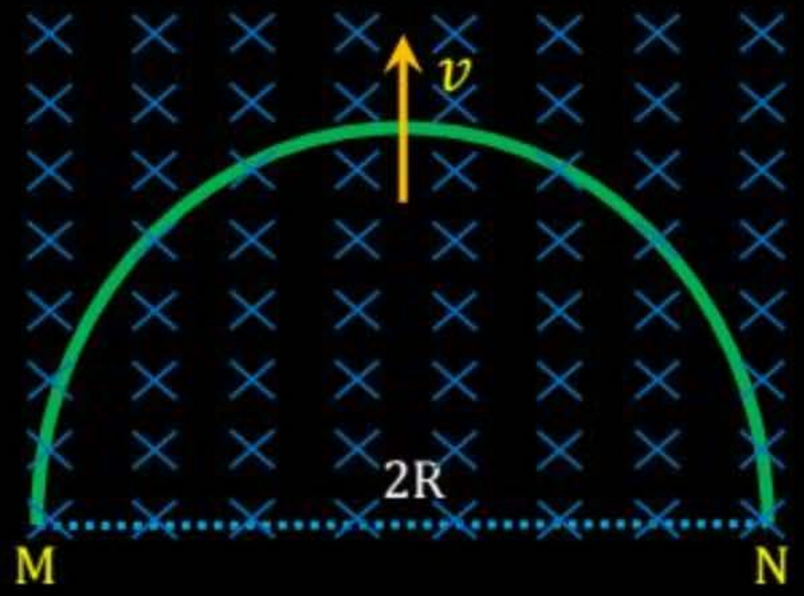
A thin semicircular conducting ring MN of radius $R = 3m$ is moving with velocity $v = 5m/s$ in uniform magnetic field $\vec{B} = (2\hat{i} + 3\hat{j} + 4\hat{k})$ T as shown. Find emf induce in the ring

$\hat{j}$ $\hat{i}$ $\hat{k}$

$\mathcal{E}.m.f = V(2R)B_z$

$= 5 \times 6 \times 4 = 120$ Volt





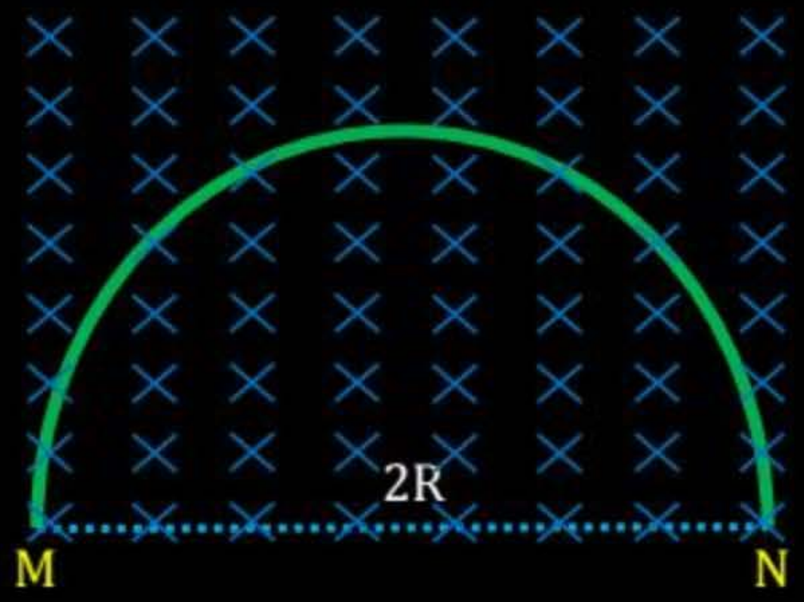
Question

A thin semicircular conducting ring MN of radius $R = 3m$ is moving with velocity $\vec{v} = (\hat{i} - 2\hat{j} + 2\hat{k})m/s$ in uniform magnetic field $\vec{B} = (2\hat{i} + 3\hat{j} + 4\hat{k})$ T as shown. Find emf induce in the ring

$$\mathcal{E}.m.f. = (\vec{V} \times \vec{B}) \cdot \vec{l}_{eq}$$

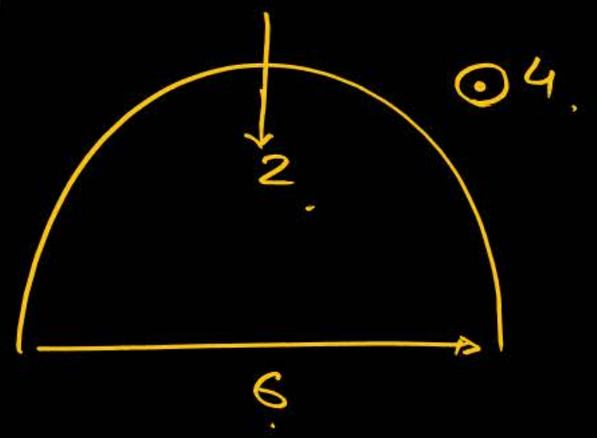
$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 2 \\ 2 & 3 & 4 \end{vmatrix} \cdot (6\hat{i})$$

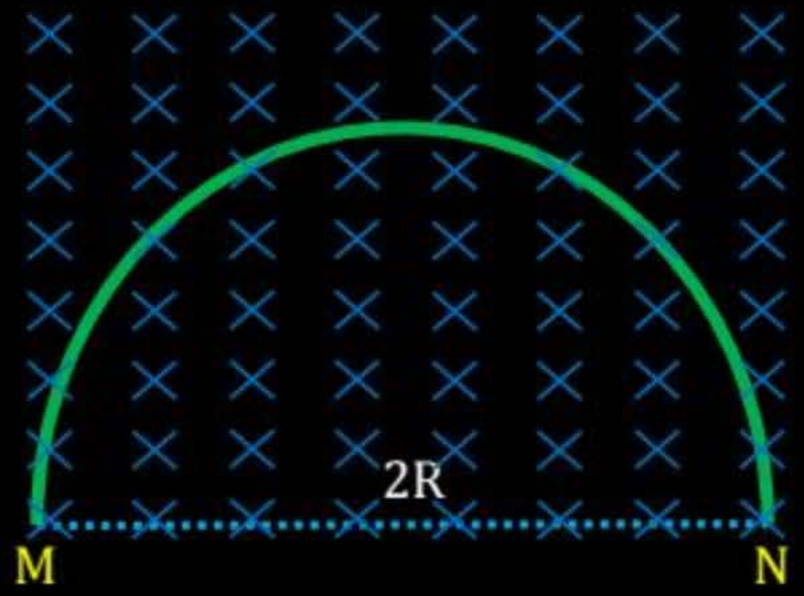
$$= \left(\hat{i}(-8-6) + \cdots\right) \cdot 6\hat{i} = -84V$$

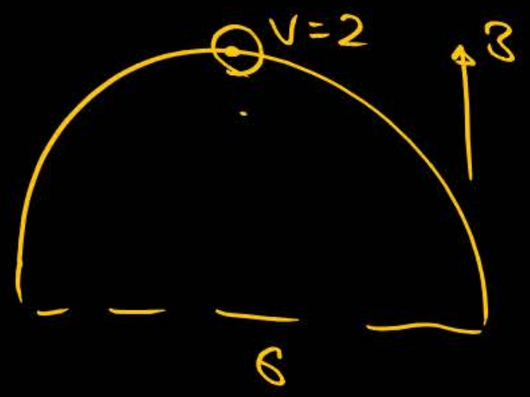


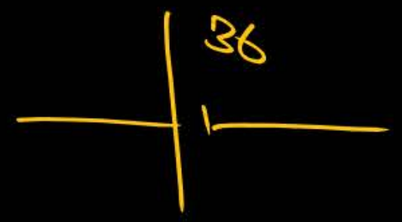
Question

A thin semicircular conducting ring MN of radius $R = 3m$ is moving with velocity $\vec{v} = (\hat{\imath} - 2\hat{\jmath} + 2\hat{k})m/s$ in uniform magnetic field $\vec{B} = (2\hat{\imath} + 3\hat{\jmath} + 4\hat{k})$ T as shown. Find emf induce in the ring

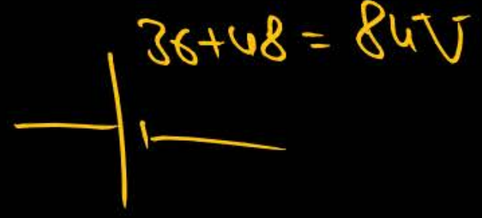








$\Rightarrow$



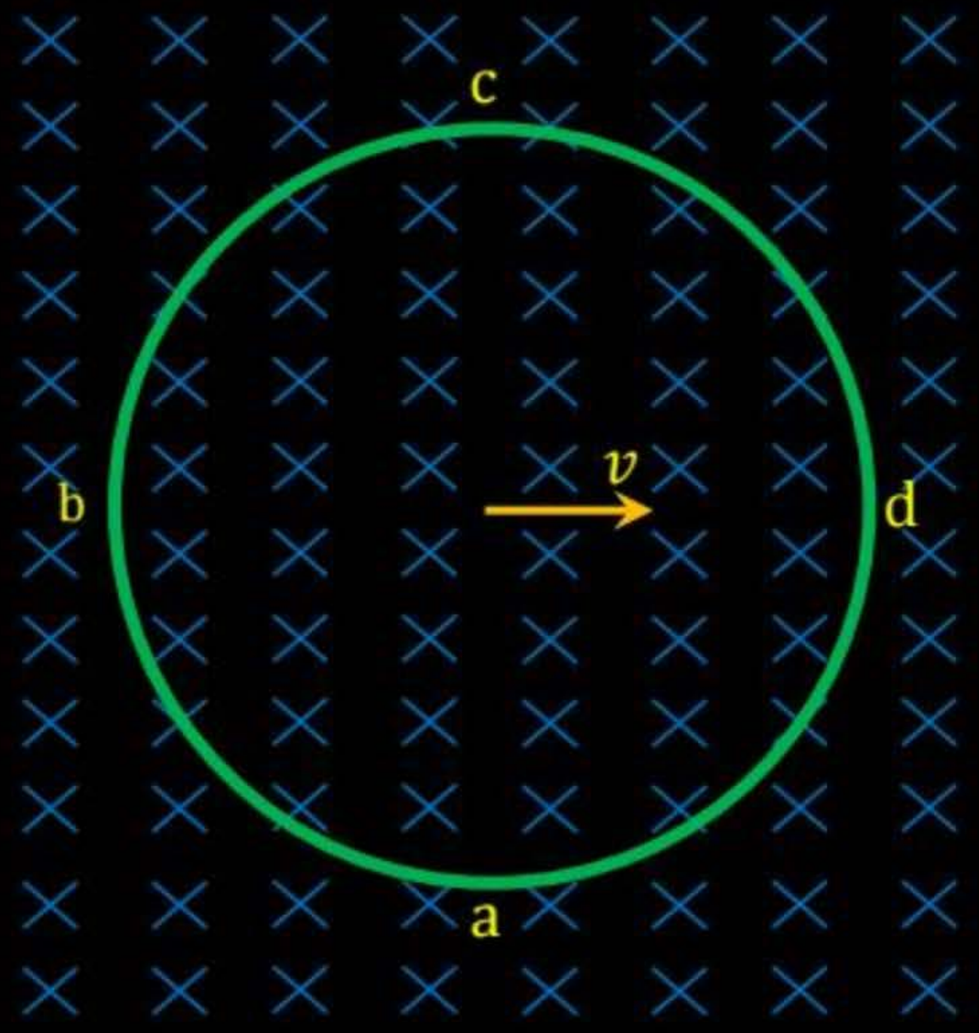
Question

A conducting wire MN is moving with velocity $\vec{v} = (\hat{\imath} - 2\hat{\jmath} + 2\hat{k})m/s$ in uniform magnetic field $\vec{B} = (2\hat{\imath} + 3\hat{\jmath} + 4\hat{k})$ T. Find emf induce in the wire. $\{M \equiv (6,8,4)\ and\ N \equiv (9,7,5)\}$

Question

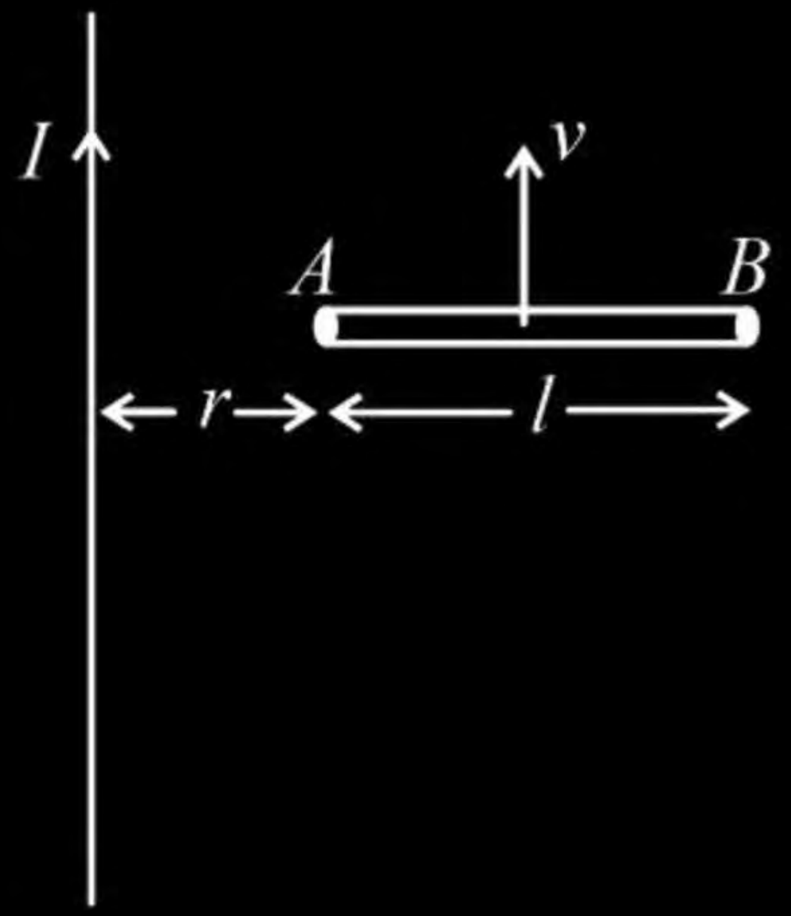
A thin conducting ring of radius R is moving with velocity v in uniform magnetic field B as shown. Find emf induce in the ring, also find emf and voltage across points

1. A and C
2. B and D
3. A and B



Question

A rod of length l is moved parallel to infinite current carrying wire with constant speed v. find potential difference developed across A & B.



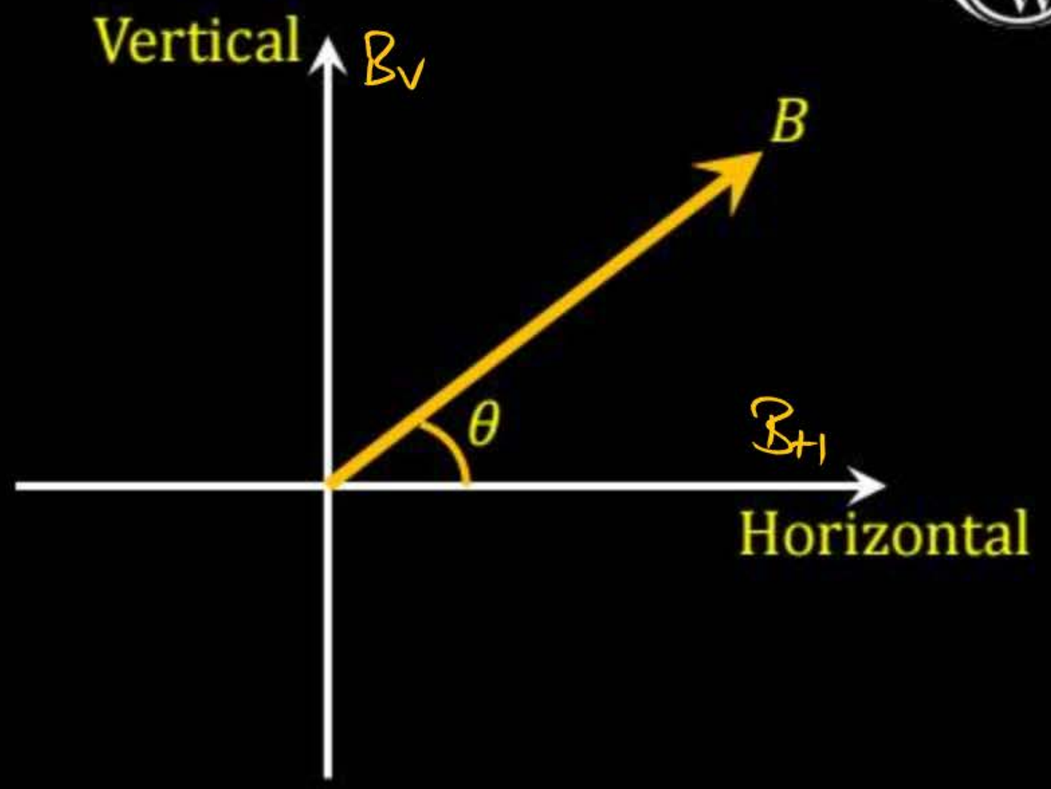
Earth Magnetic Field

Earth's magnetic field is at angle θ with horizontal. This angle is known as angle of dip

Direction of horizontal component of earth is from $\boxed{\text{South to North}}$

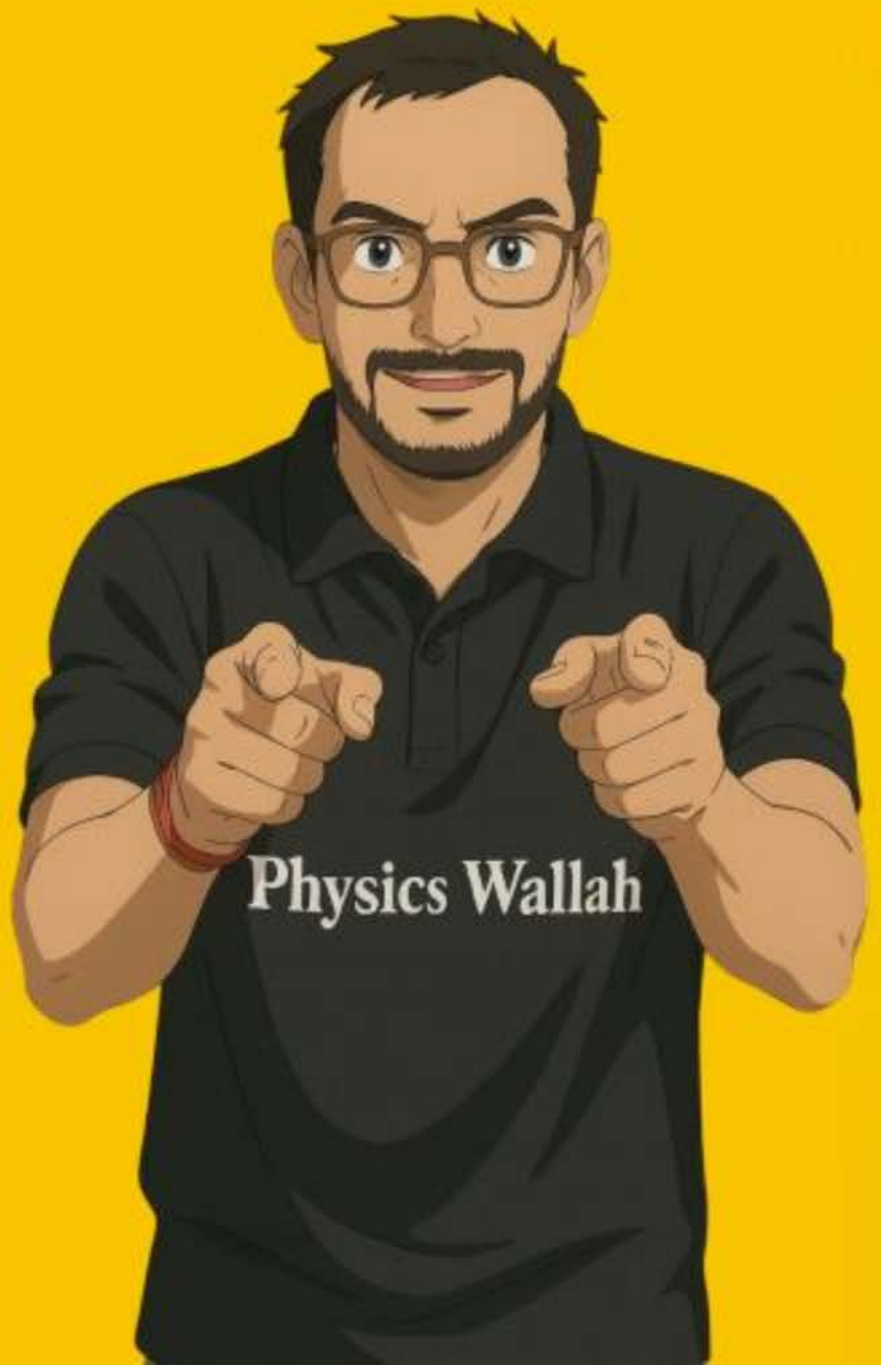
$B_H = B\cos\theta$, $B_V = B\sin\theta$

$\tan\theta = \frac{B_V}{B_H}$, $B = \sqrt{B_H^2 + B_V^2}$



Question

An aircraft is flying with 500m/s with 40 m wingspan in East. The earth's magnetic field in its plane is 2×10^{-4} T & angle of dip is 30°. Find emf induced between two ends.



THANK YOU BAWAL BACCHA PARTY